Lecture 20 (11/2)

Residual plots and feature maps

Today:

1) Residual plots to identify interactions and other model deficiencies
2) Feature maps: nonlinear models that are linear in the parameters

Residual plots

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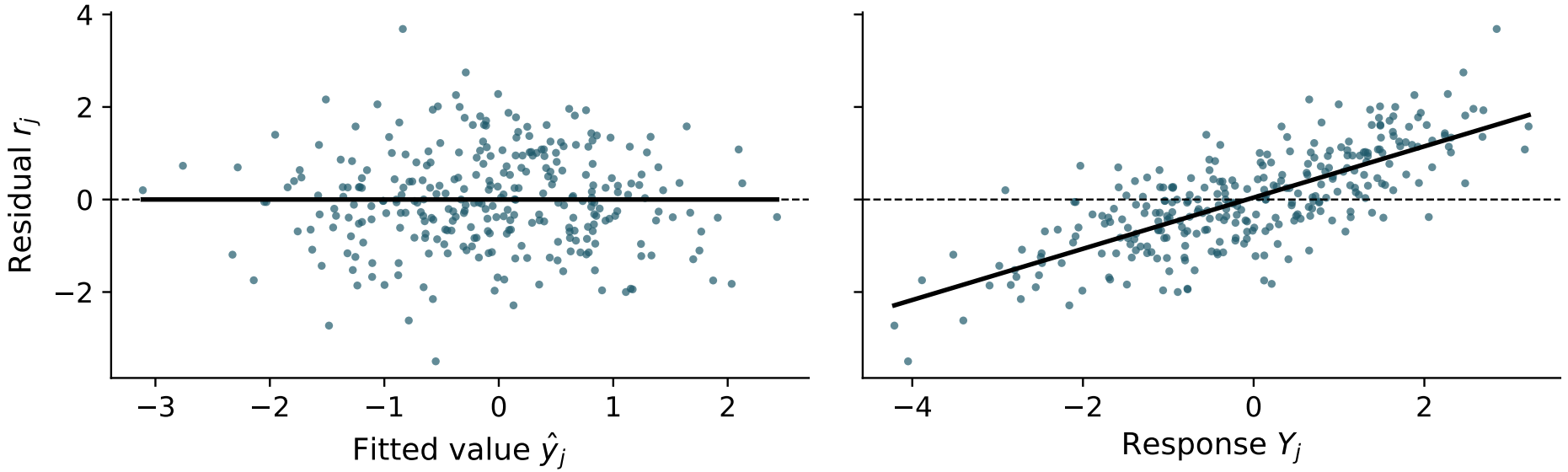
(r_j is computed from the data; ϵ_j in the model cannot be observed.)

If the model is right, r_j is approximately Normal$(0, \sigma_\epsilon^2)$ *whatever the values of the predictors*. A residual plot is meant to visualize the r_j.

Residuals vs. fitted values and vs. Y

Consider

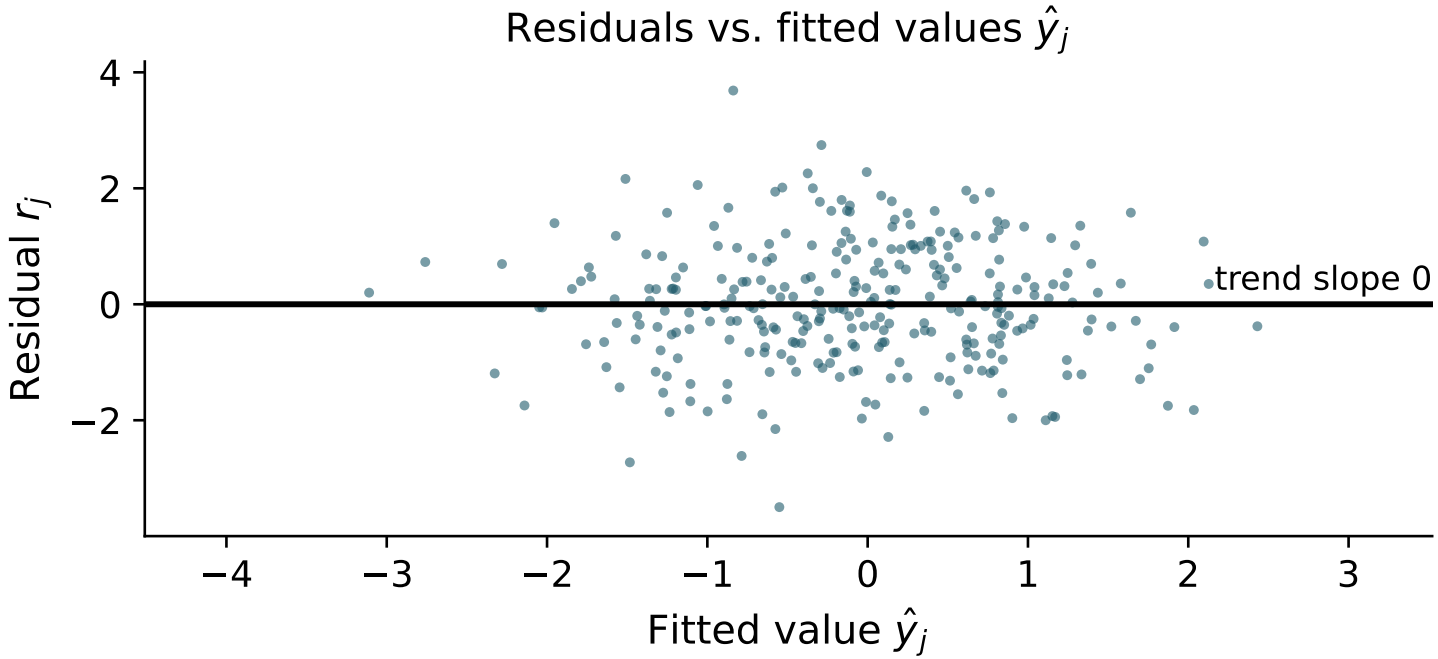
$$Y = 0.2X_1 + 20X_2 + \epsilon, \quad \sigma_\epsilon = 1$$



Question

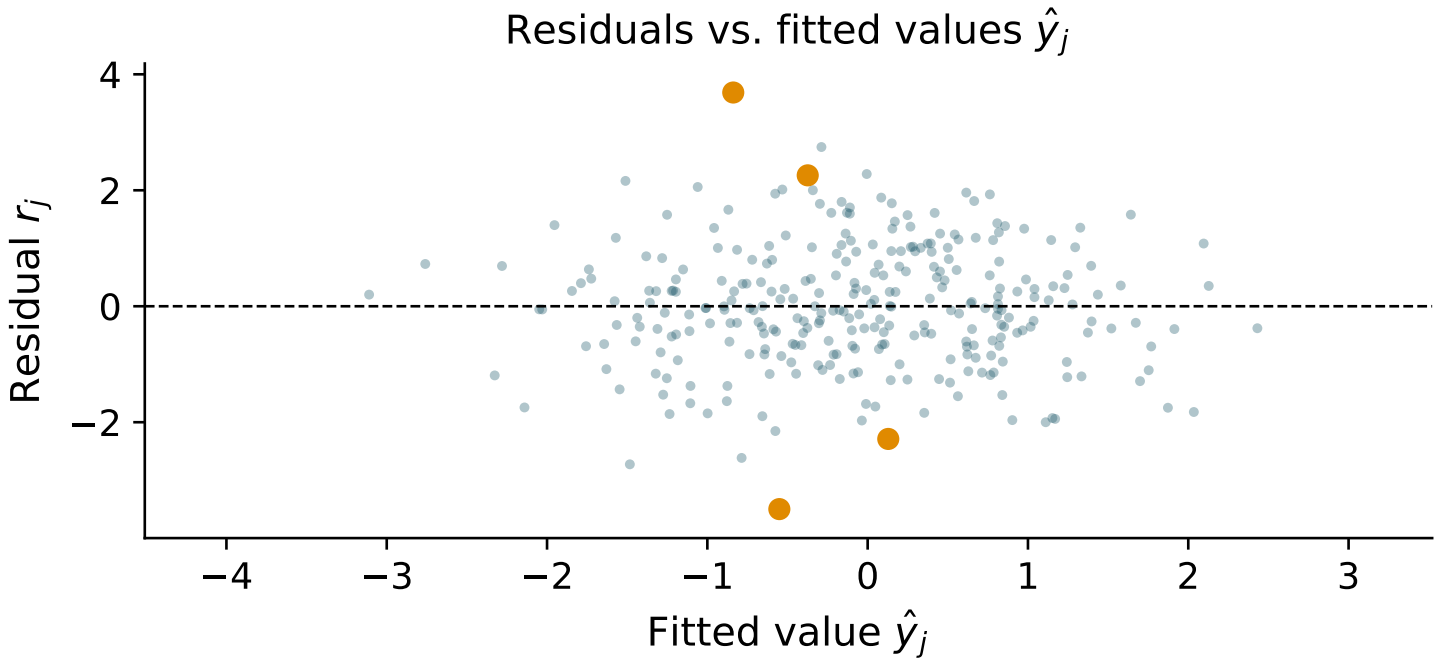
Why is there a trend on the right plot?

Why plot $\hat{y}_i$ on x axis



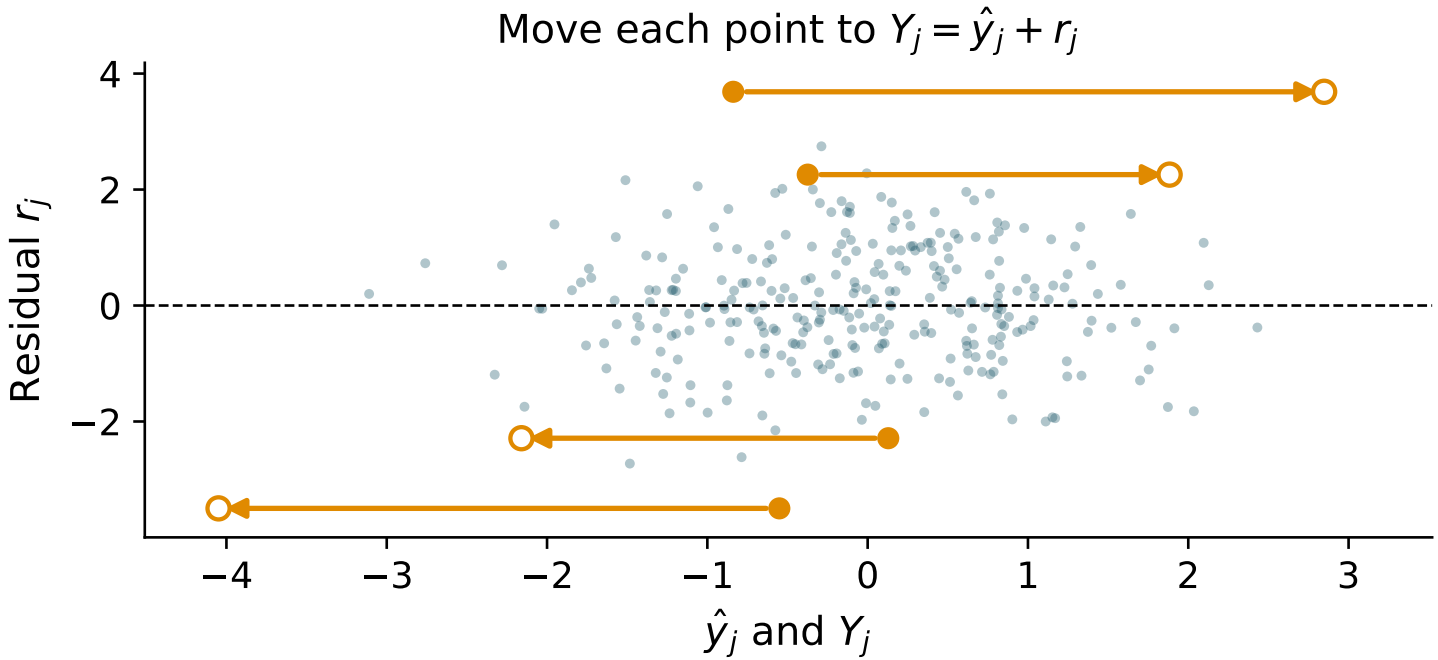
1. If the model is right, r_j has the same distribution whatever the predictors are, so vs. $\hat{y}_j$ we expect a symmetric spread in the vertical direction.

Why plot $\hat{y}_i$ on x axis



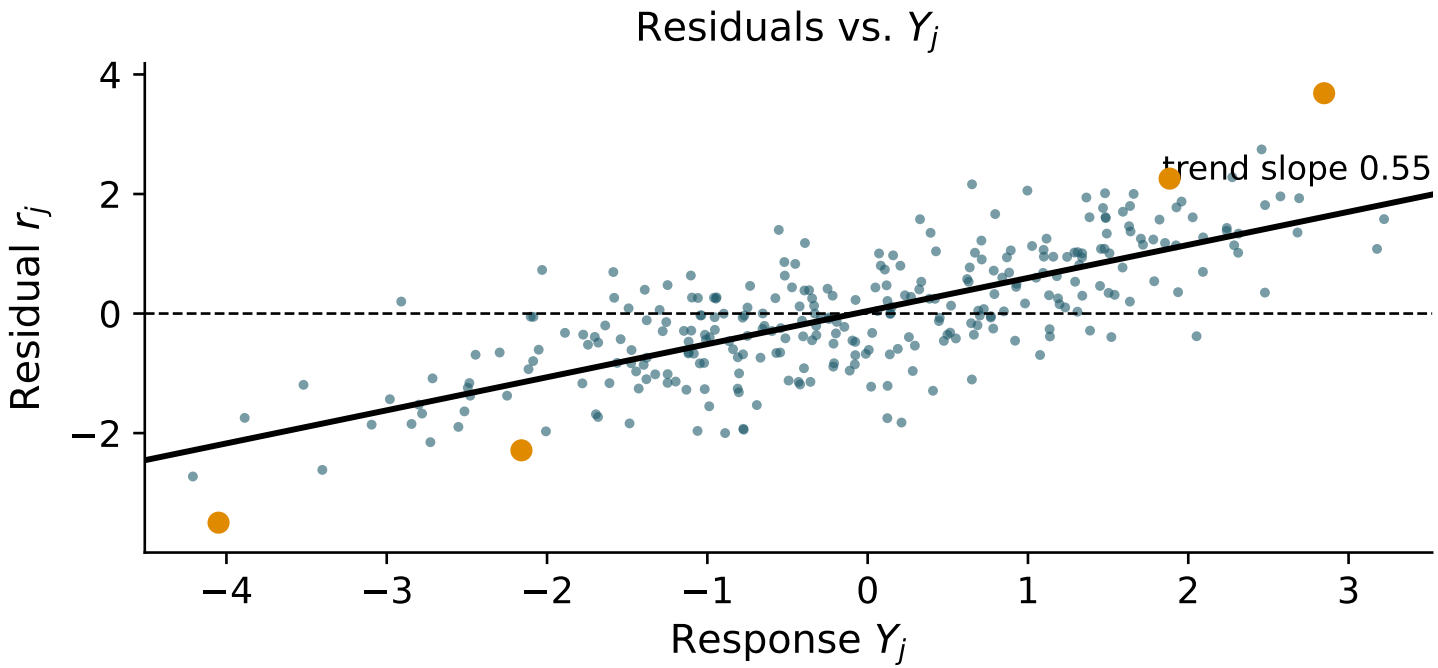
2. Pick out a few points with large residuals.

Why plot $\hat{y}_j$ on x axis



3. Since $Y_j = \hat{y}_j + r_j$, putting Y_j on the horizontal axis shifts each point sideways by the corresponding residual.

Why plot $\hat{y}_i$ on x axis



4. Every point shifts this way, so the cloud tilts even though the model is correct.

Example

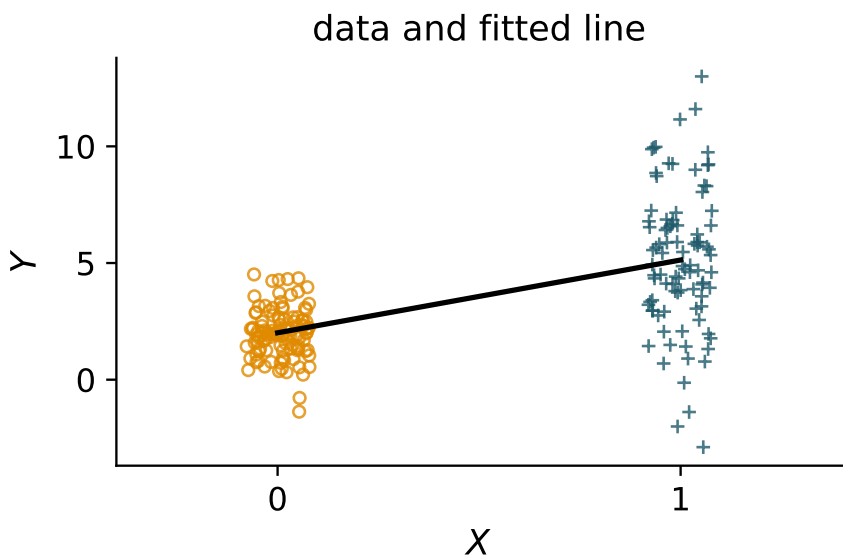
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$$Y = 2 + 3X + \epsilon(1 + 2X)$$

Below is the data with the fitted line.

Question

What will the residual plot look like when the data is fit to a linear regression model?



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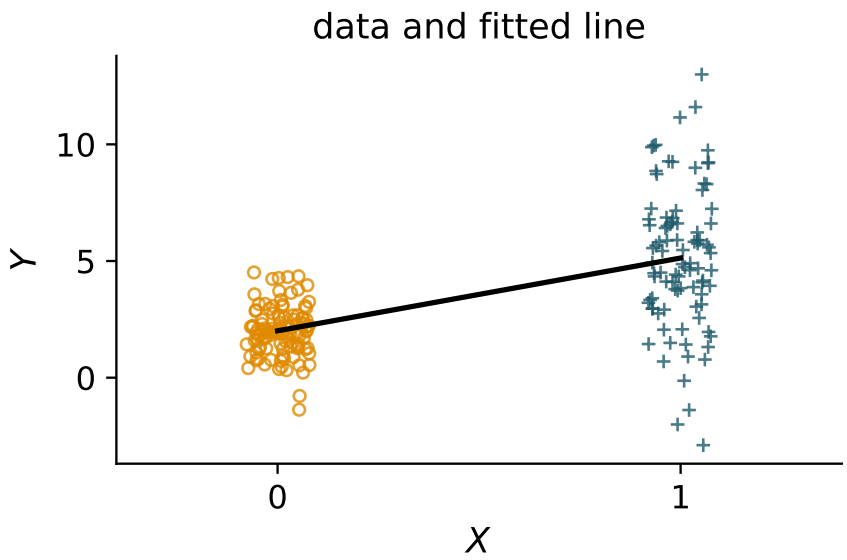
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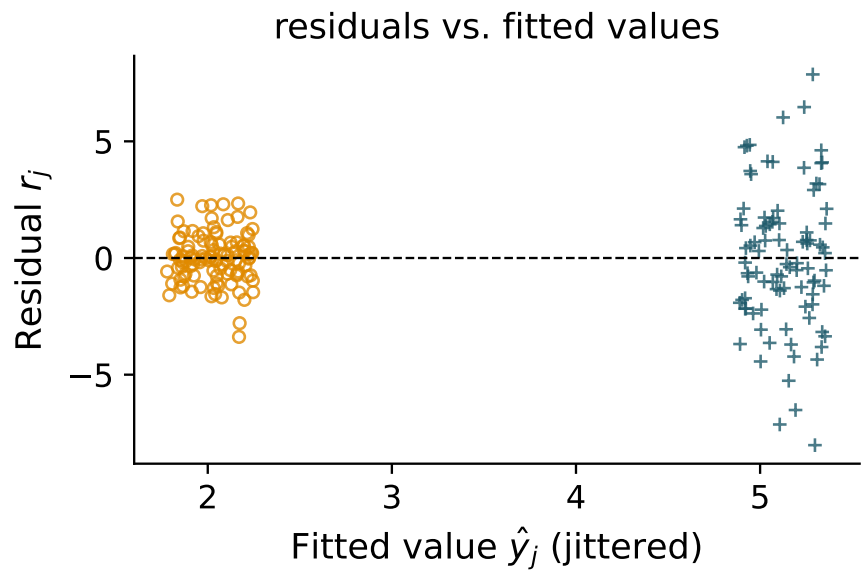
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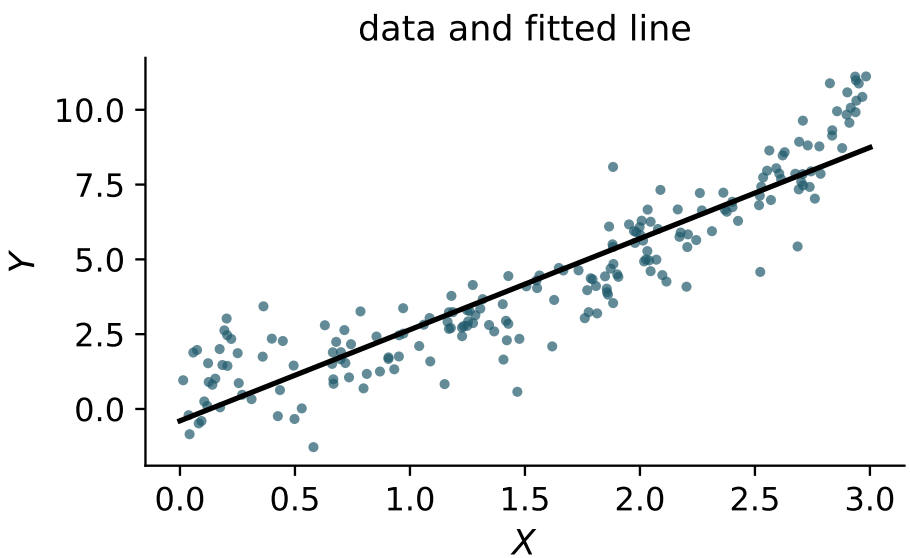
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where X is between 0 and 3. Below is the data with the fitted line.

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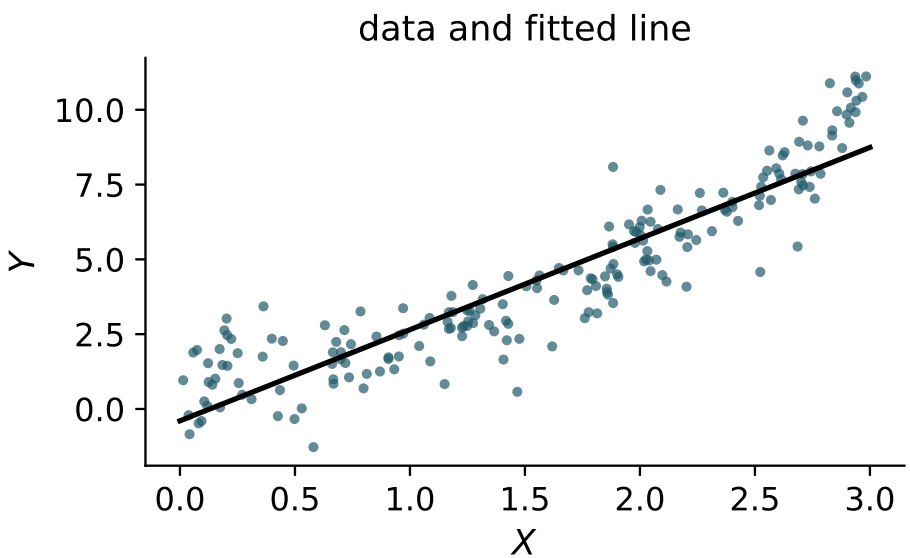
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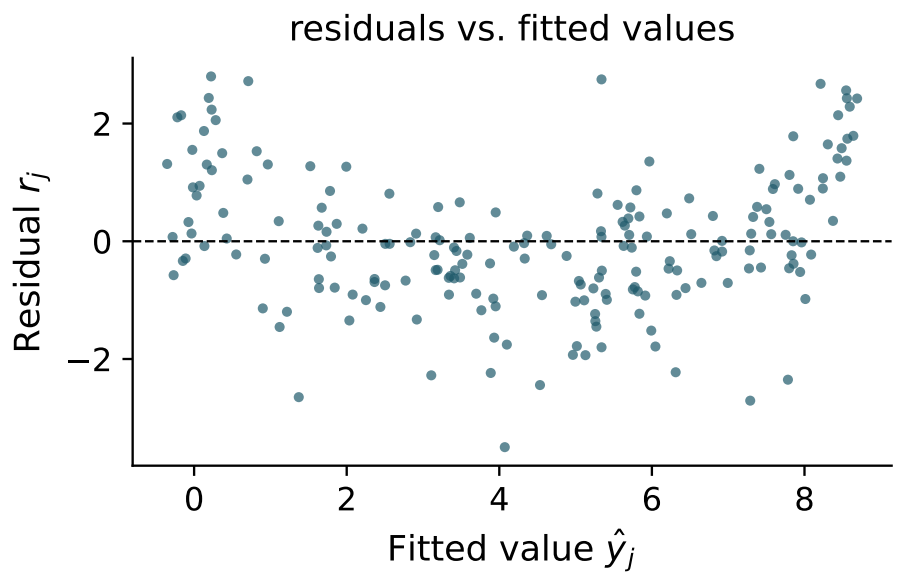
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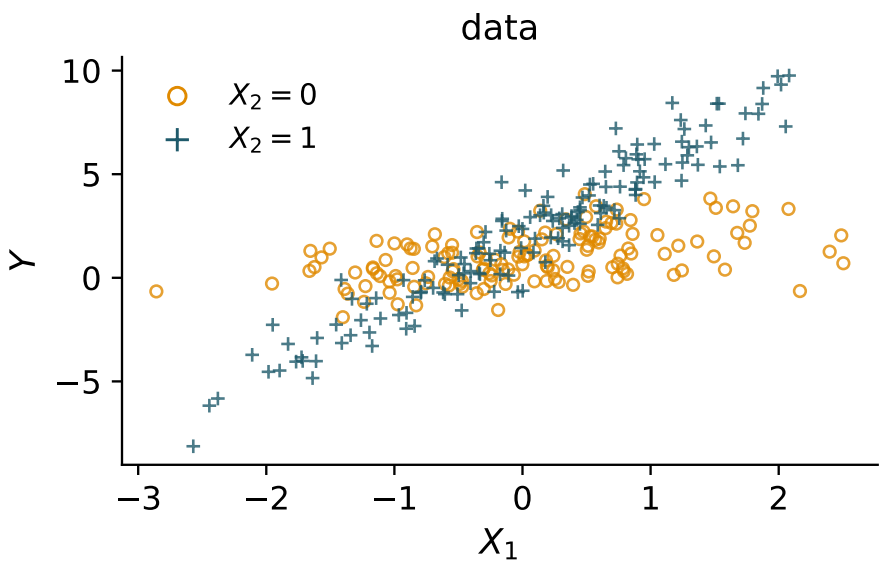
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$$Y = 1 + 0.5X_1 + X_2 + 3X_1X_2 + \epsilon$$

where X_2 is a binary predictor. Below is the data.

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Sketch the residual plot when this model is fitted without the interaction term.



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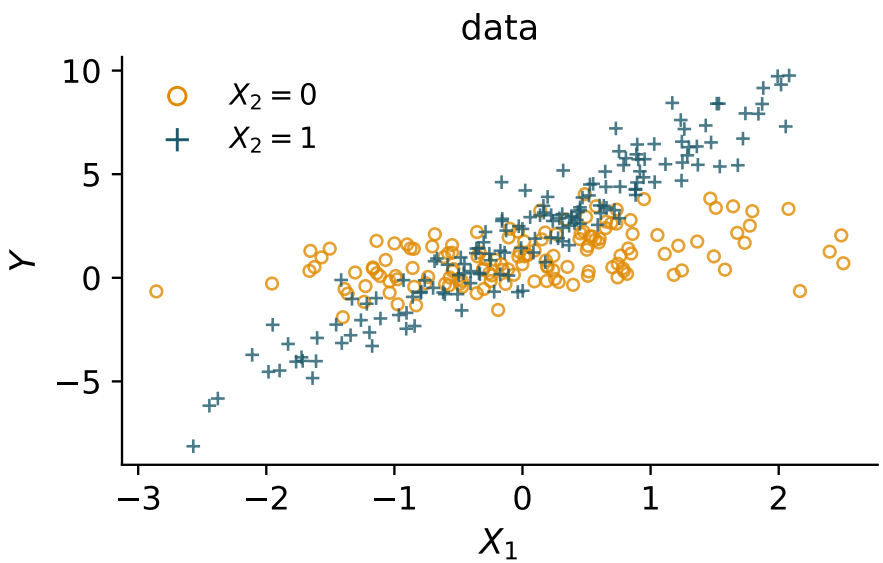
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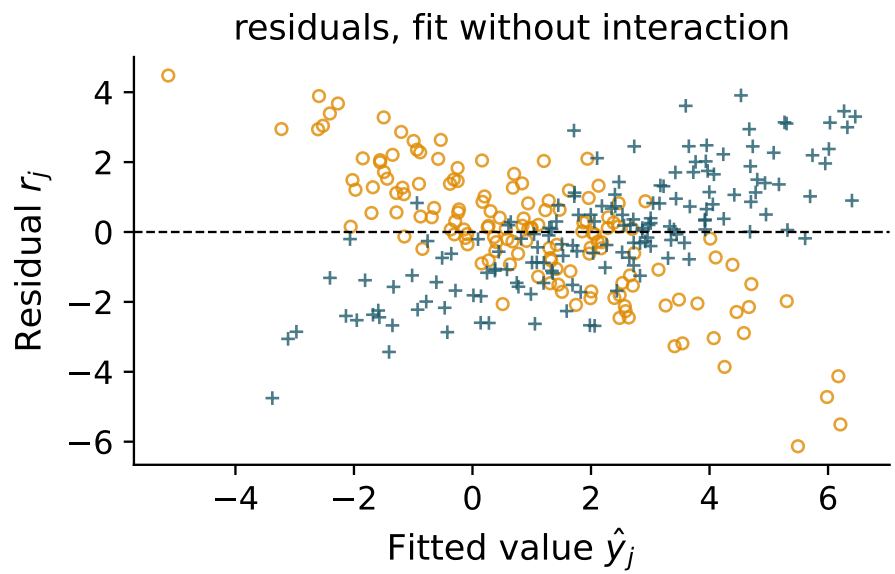
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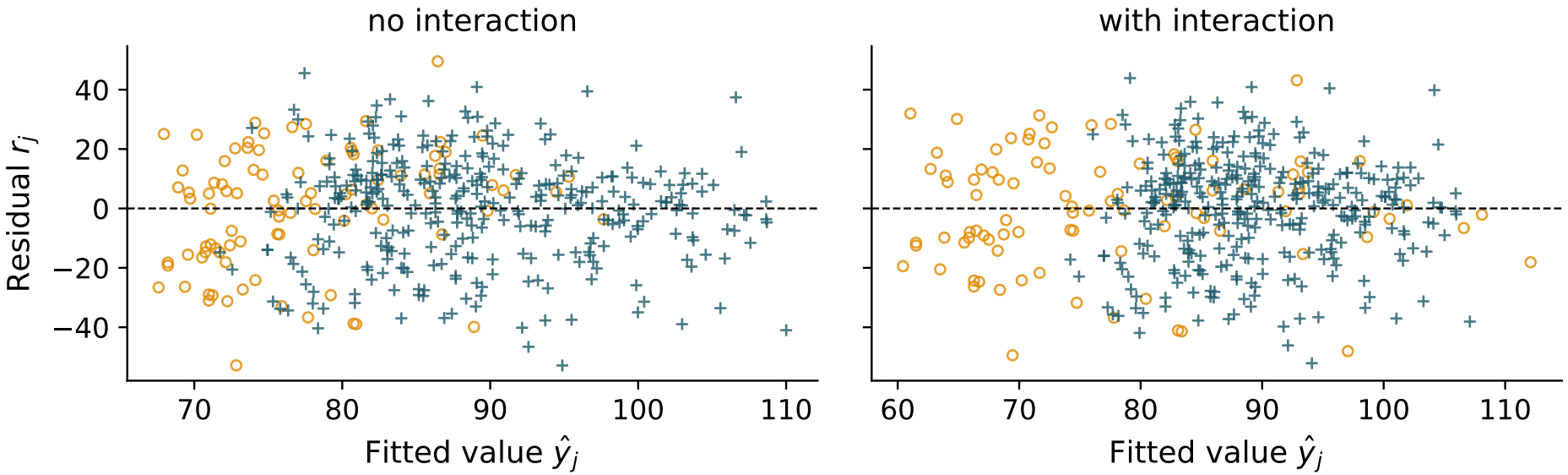
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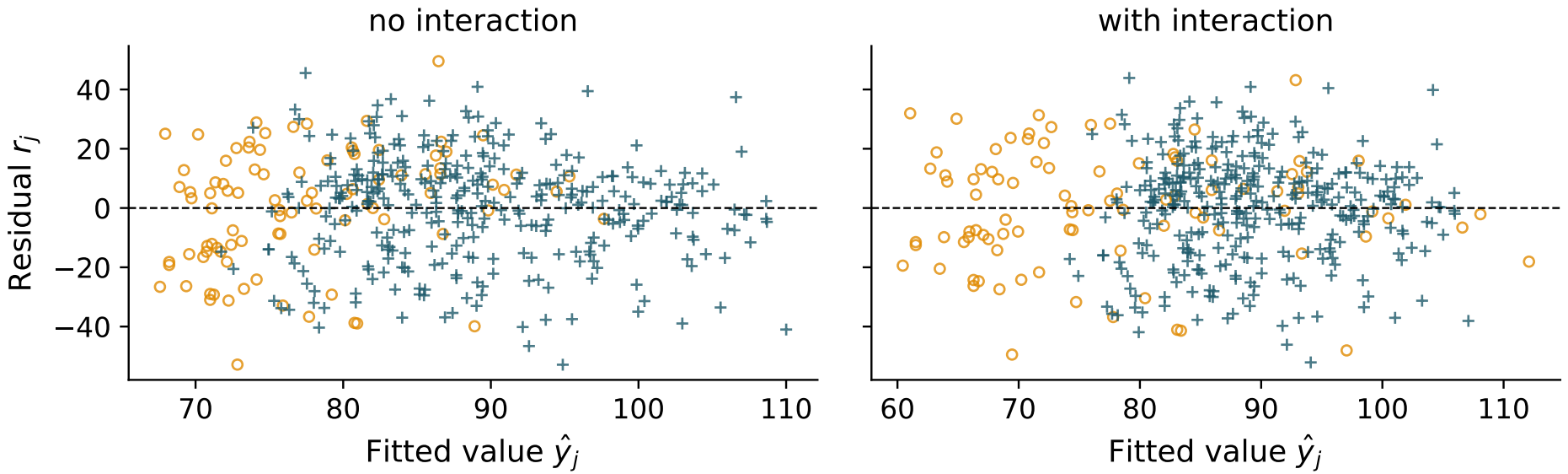




Test scores: residuals vs. fitted values



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Question

The missing interaction is not evident from the residual plot, but it is statistically significant ($p = 0.003$). What does this suggest about how R^2 changes when adding the residual?

Example: advertising budgets and sales

Sales of a product in 200 markets (from *An Introduction to Statistical Learning*): Y is sales in thousands of units, and X_{TV}, X_{radio} are the TV and radio advertising budgets in \$1000s. Without an interaction:

Dep. Variable:		sales	R-squared:			0.897
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.9211	0.294	9.919	0.000	2.340	3.502
TV	0.0458	0.001	32.909	0.000	0.043	0.048
radio	0.1880	0.008	23.382	0.000	0.172	0.204

With an interaction, and both budgets centered at their means (147.0 and 23.3):

Dep. Variable:		sales	R-squared:			0.968
	coef	std err	t	P>|t|	[0.025	0.975]
const	13.9470	0.067	208.737	0.000	13.815	14.079
TV_c	0.0444	0.001	56.673	0.000	0.043	0.046
radio_c	0.1886	0.005	41.806	0.000	0.180	0.198
TV_radio_c	0.0011	5.24e-05	20.727	0.000	0.001	0.001

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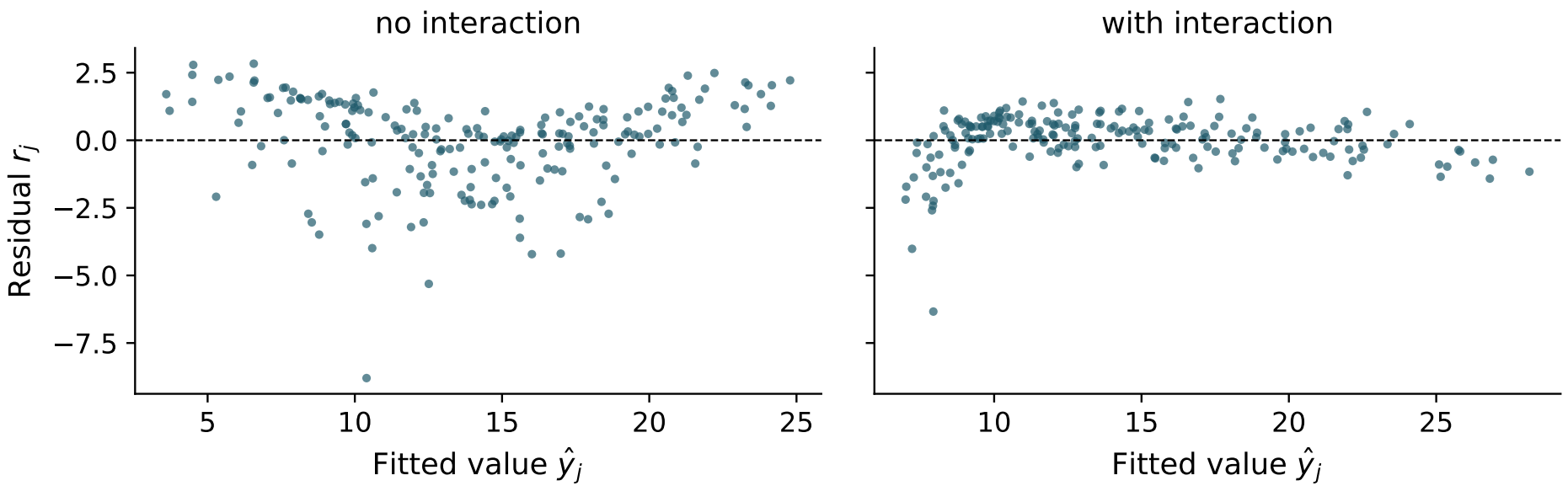
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How does the effect of an extra \$1000 of radio advertising on sales depend on the TV budget?

Advertising: residuals without vs. with interaction



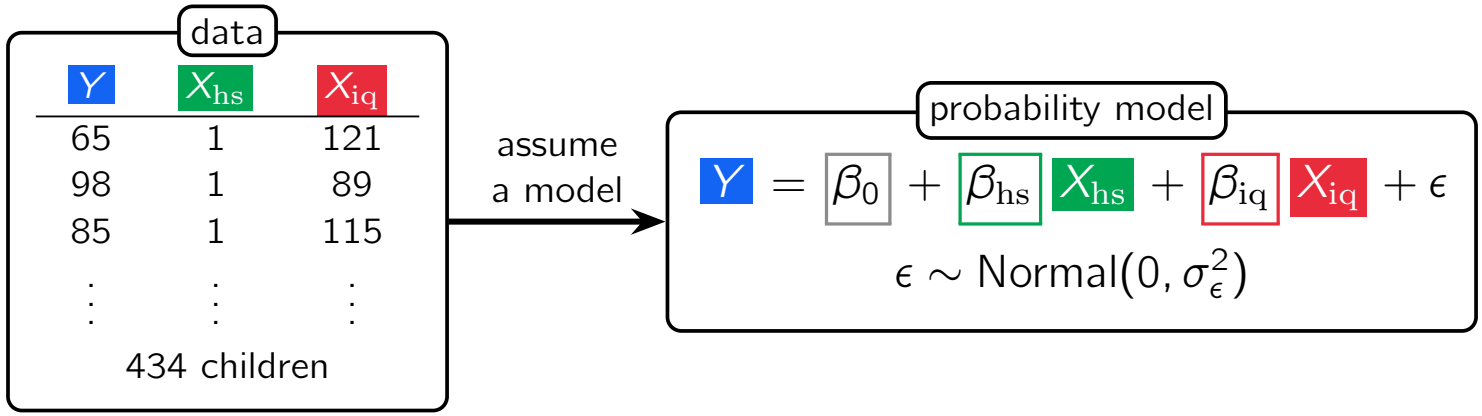
What is statistics? Recap

data

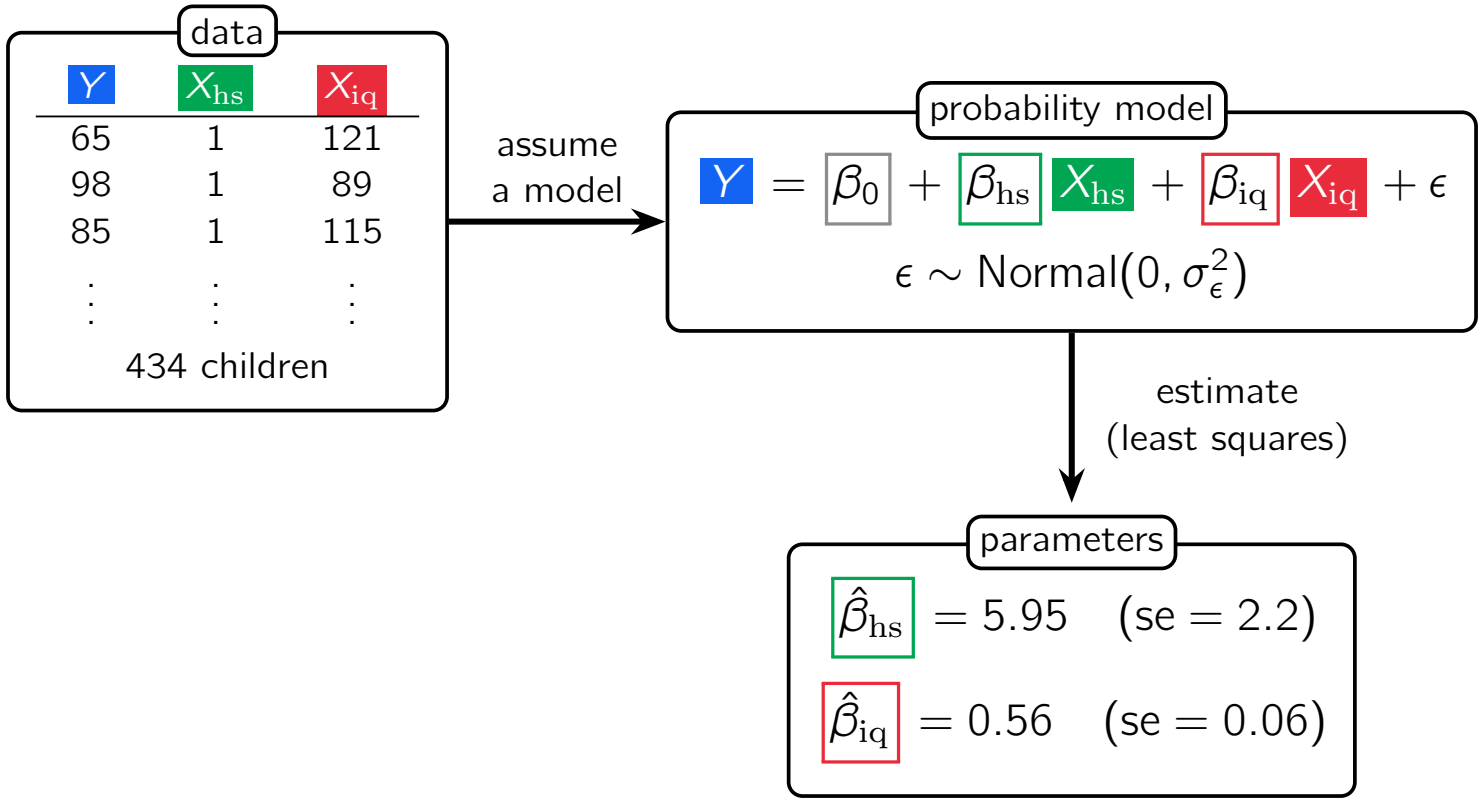
Y	X_{hs}	X_{iq}
65	1	121
98	1	89
85	1	115
⋮	⋮	⋮

434 children

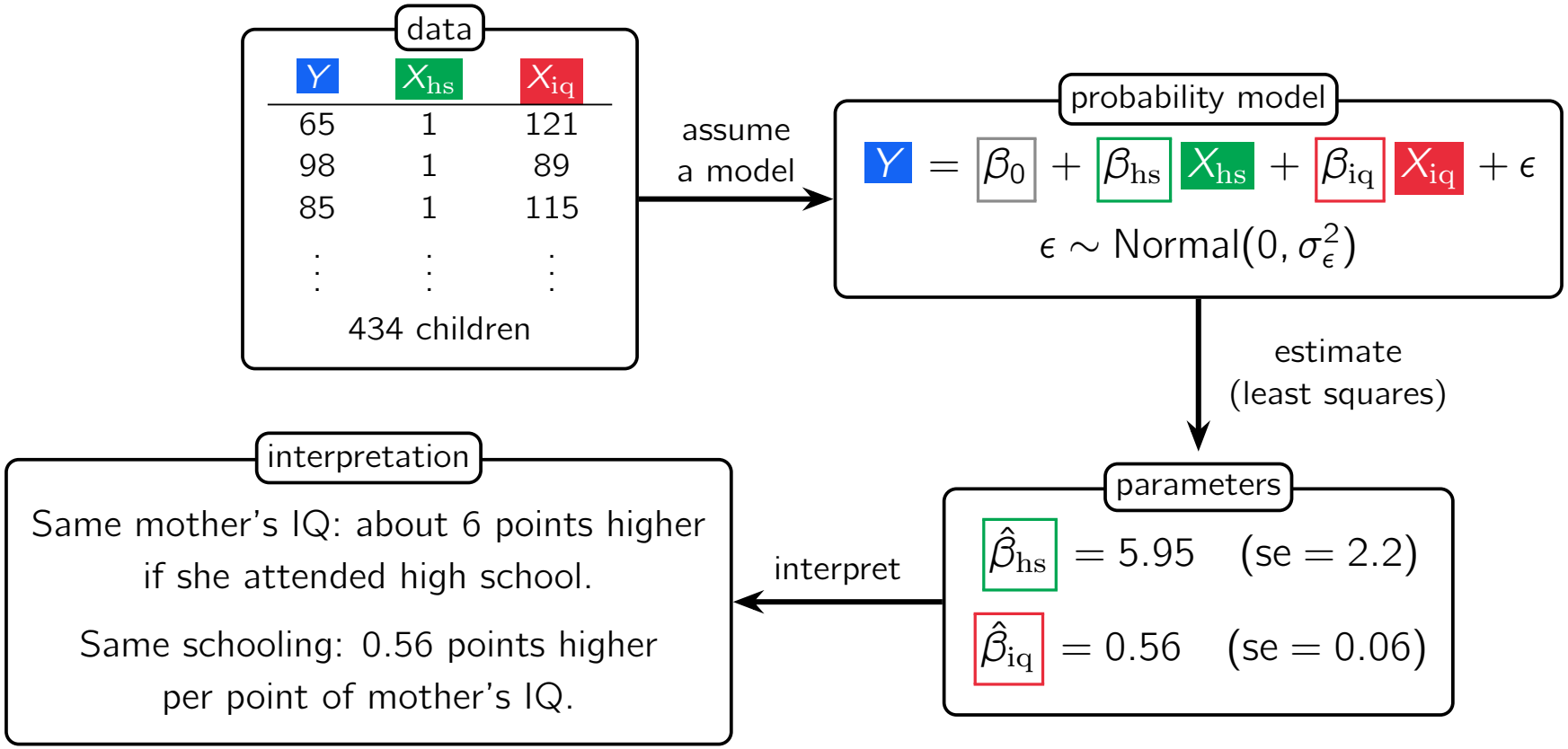
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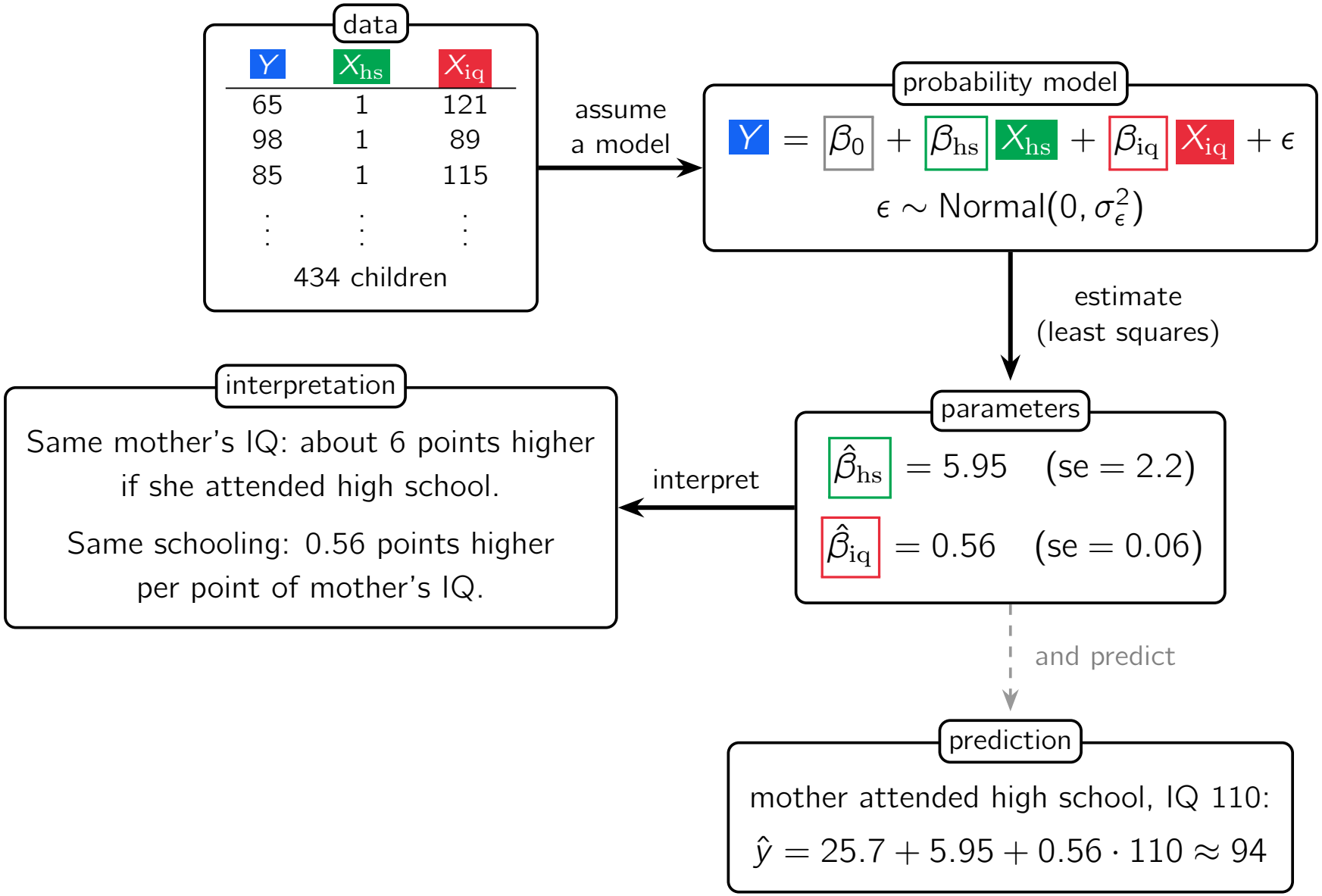
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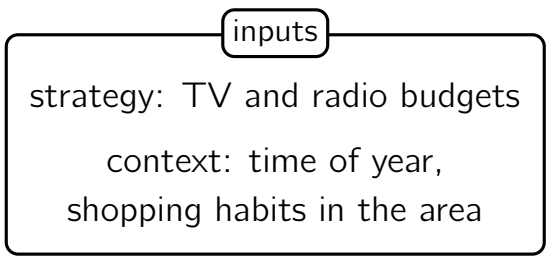


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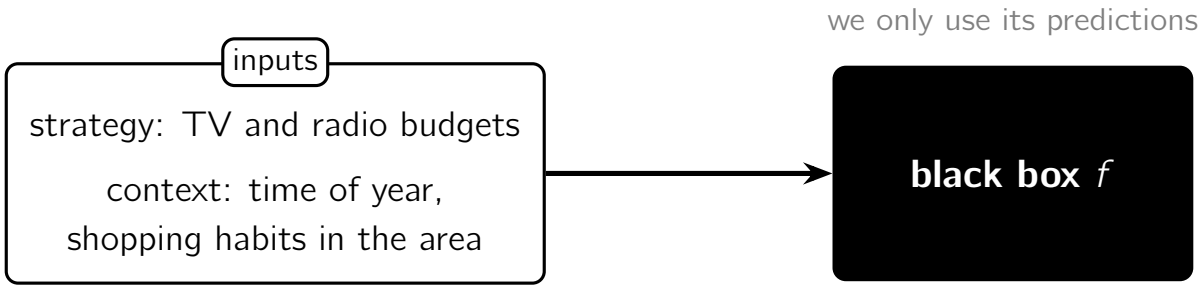
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Advertising again, but now we do not care what the parameters mean. We want to *predict* sales for any strategy, and then pick the best strategy.



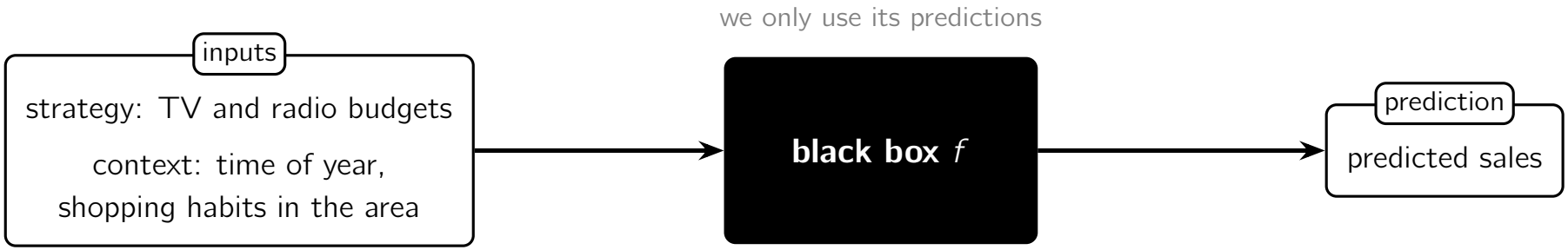
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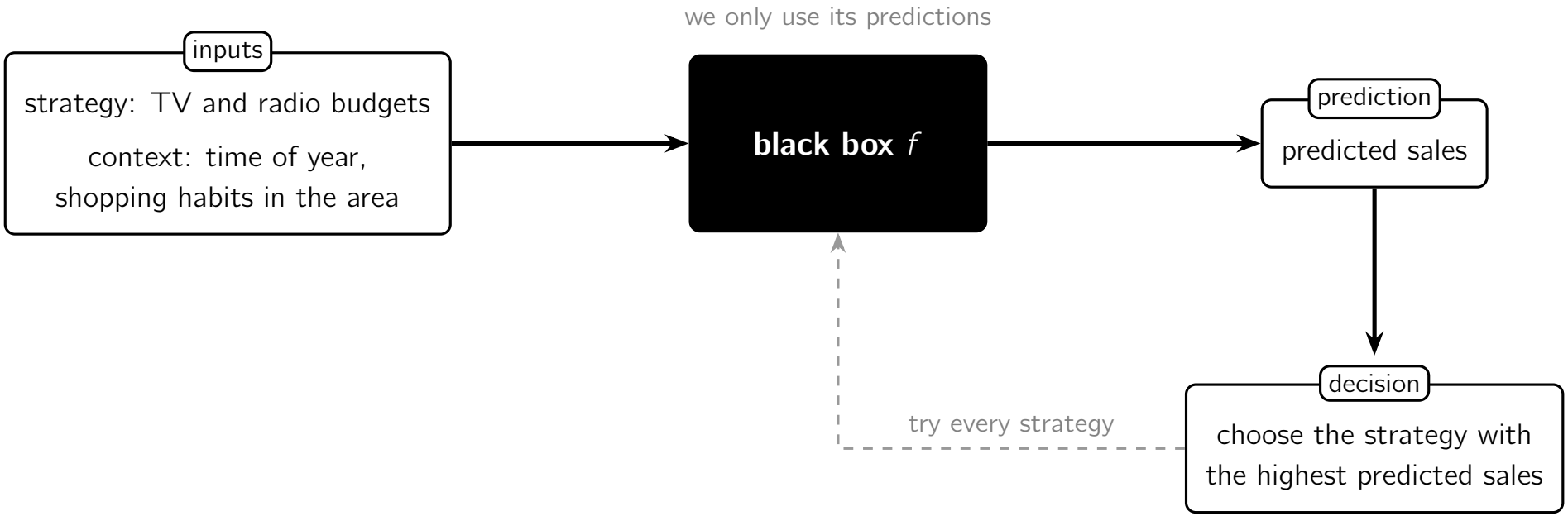
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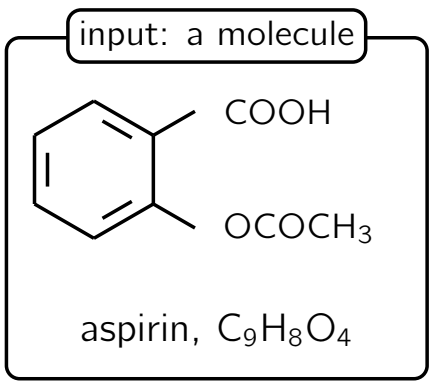
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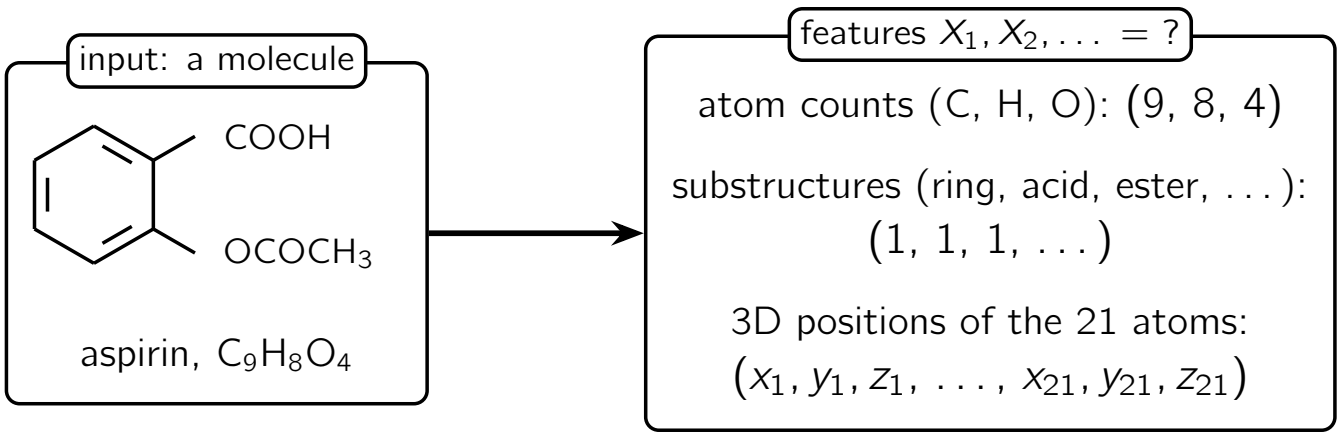
Example: drug discovery

From a molecule, predict how strongly it binds a target protein (its binding energy). The input is a chemical structure, and the predictors X_i are up to us.



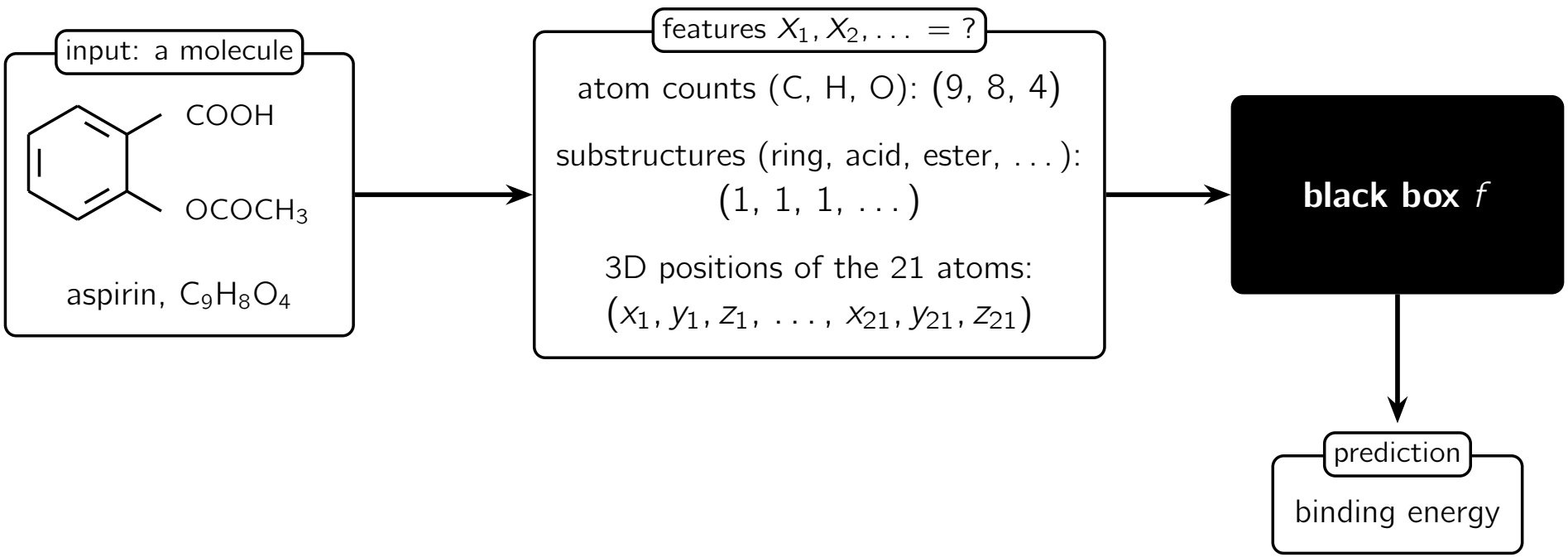
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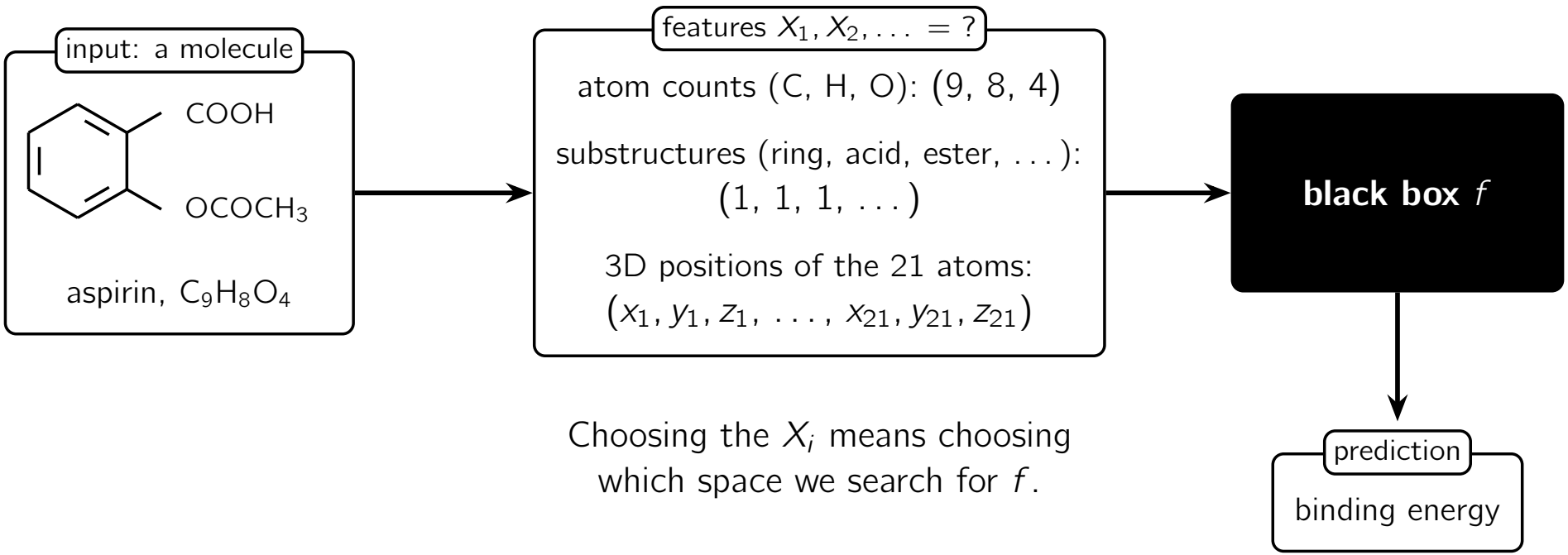
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Choosing the X_i means choosing which space we search for f.

What is the difference between statistics and ML?

Proposed answer	Problem with it
Statistics is about inference, ML is about prediction	Inference is a form of prediction
ML finds parameters with gradient descent	A consequence of ML models being more complex; the models themselves are of the same kind
Statistics quantifies uncertainty	ML can also quantify uncertainty (e.g. Gaussian processes)
ML models have more parameters than data points	Not always true
In statistics the parameters have interpretations	Also a consequence of model complexity

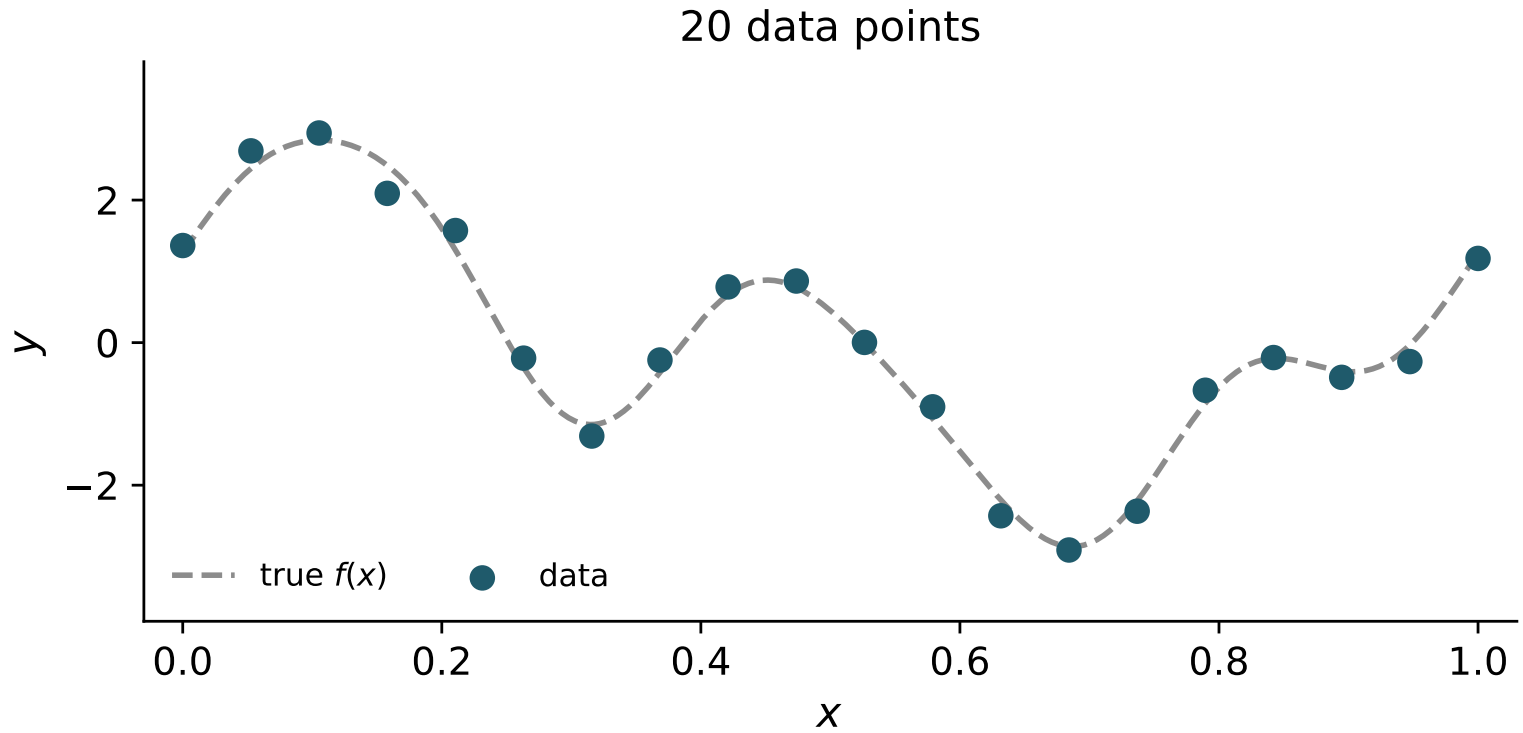
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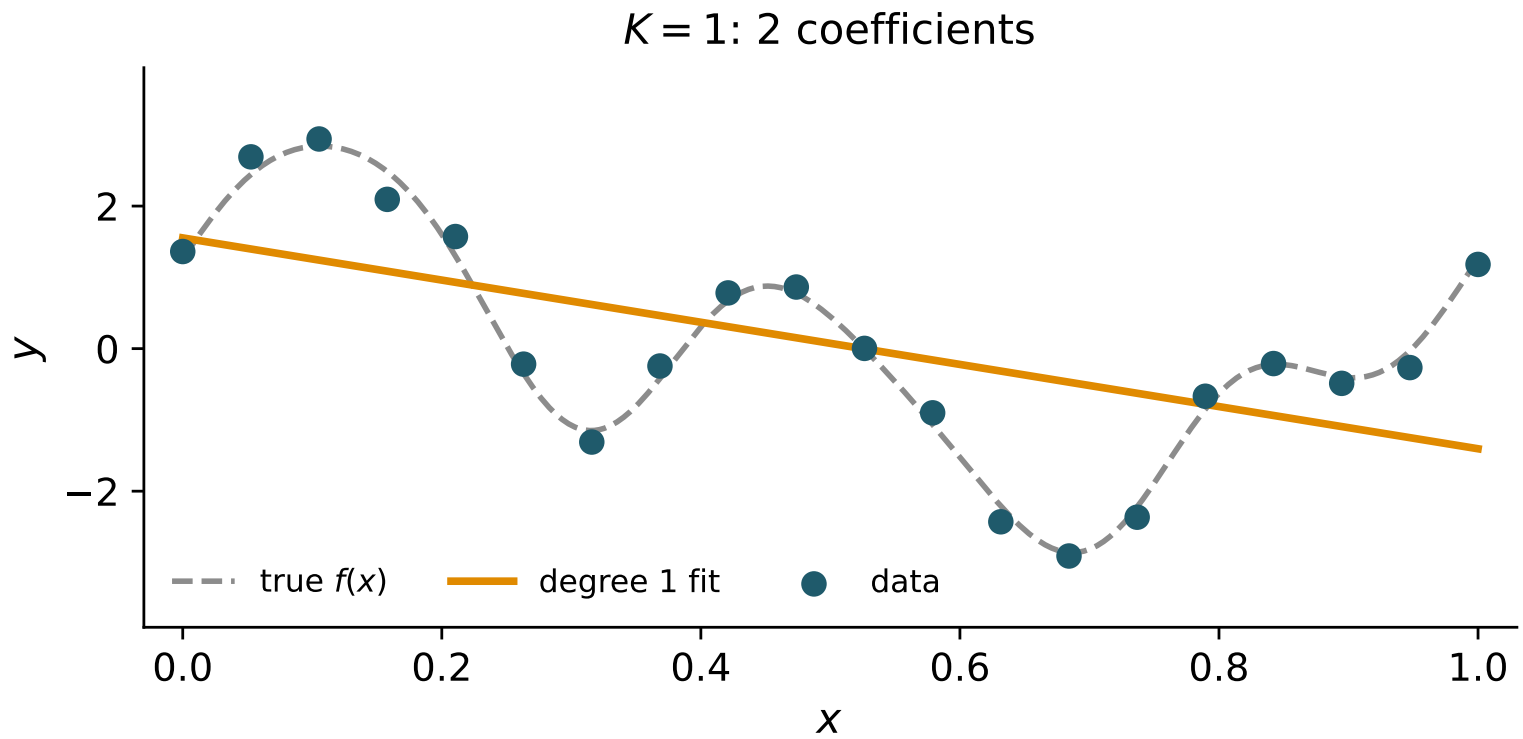
My preferred answer: statistics and ML are different regimes of the same thing, statistical learning. We can understand the two regimes with a simple example: interpolation.

Further reading: L. Breiman (2001), Statistical modeling: the two cultures, *Statistical Science* 16(3), 199–231.

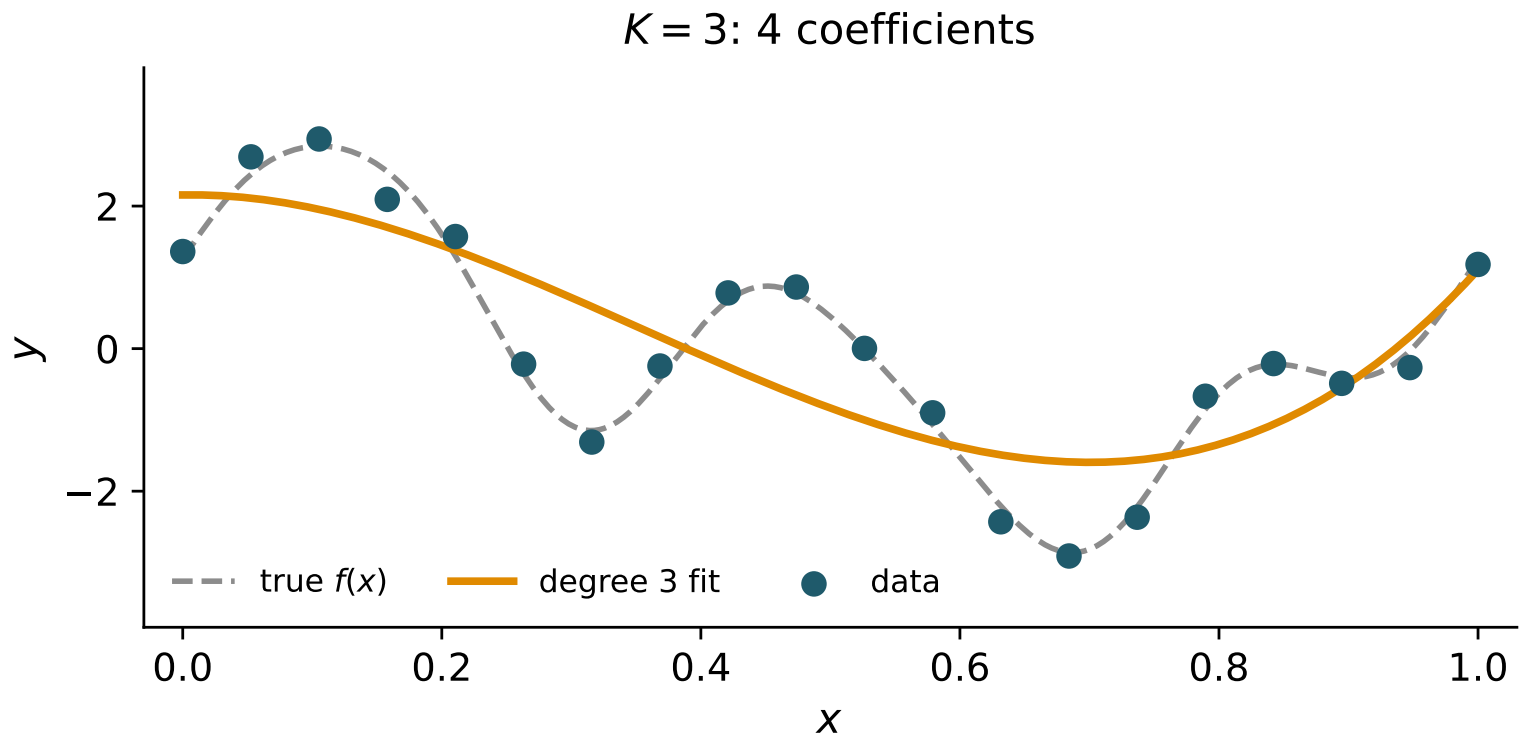
Example: polynomials fitted to a random function



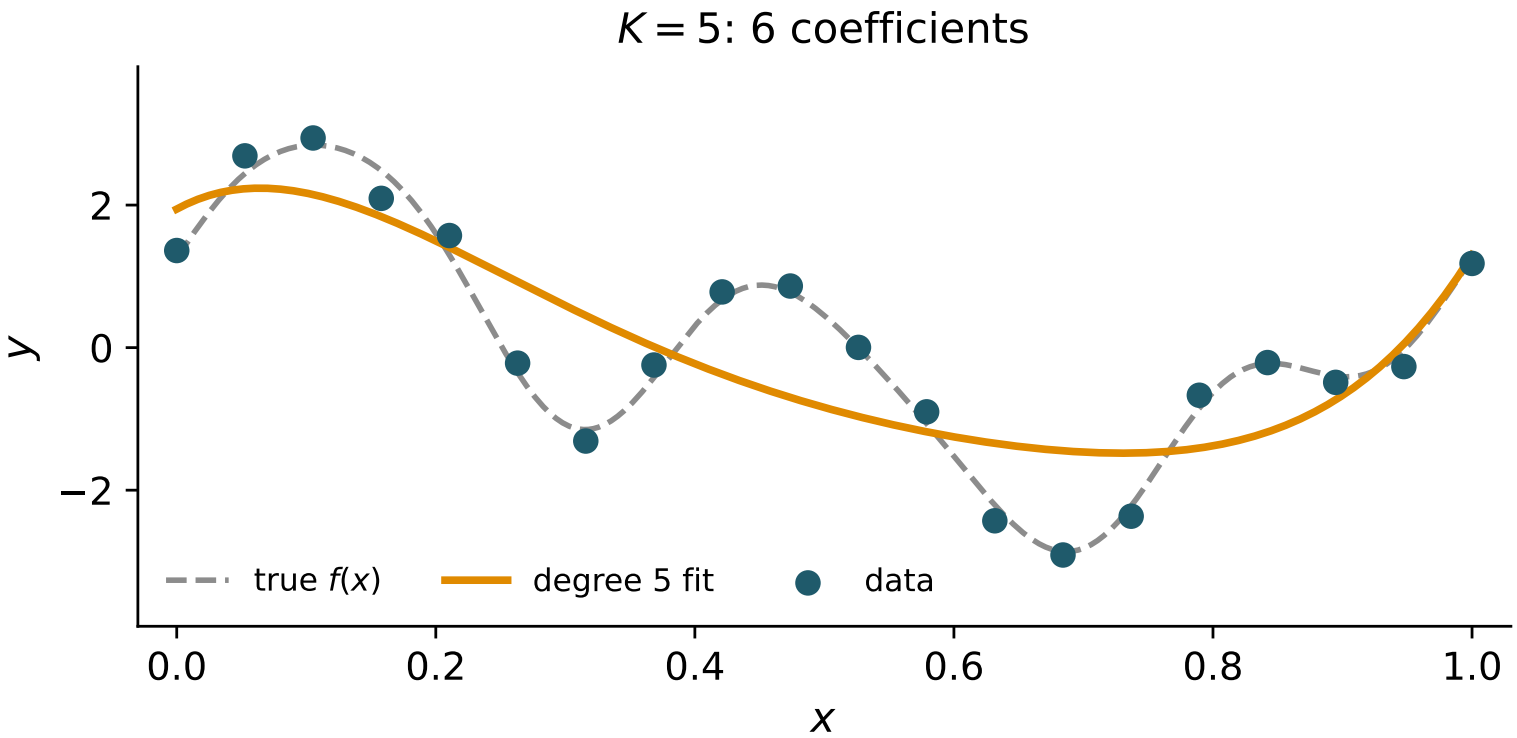
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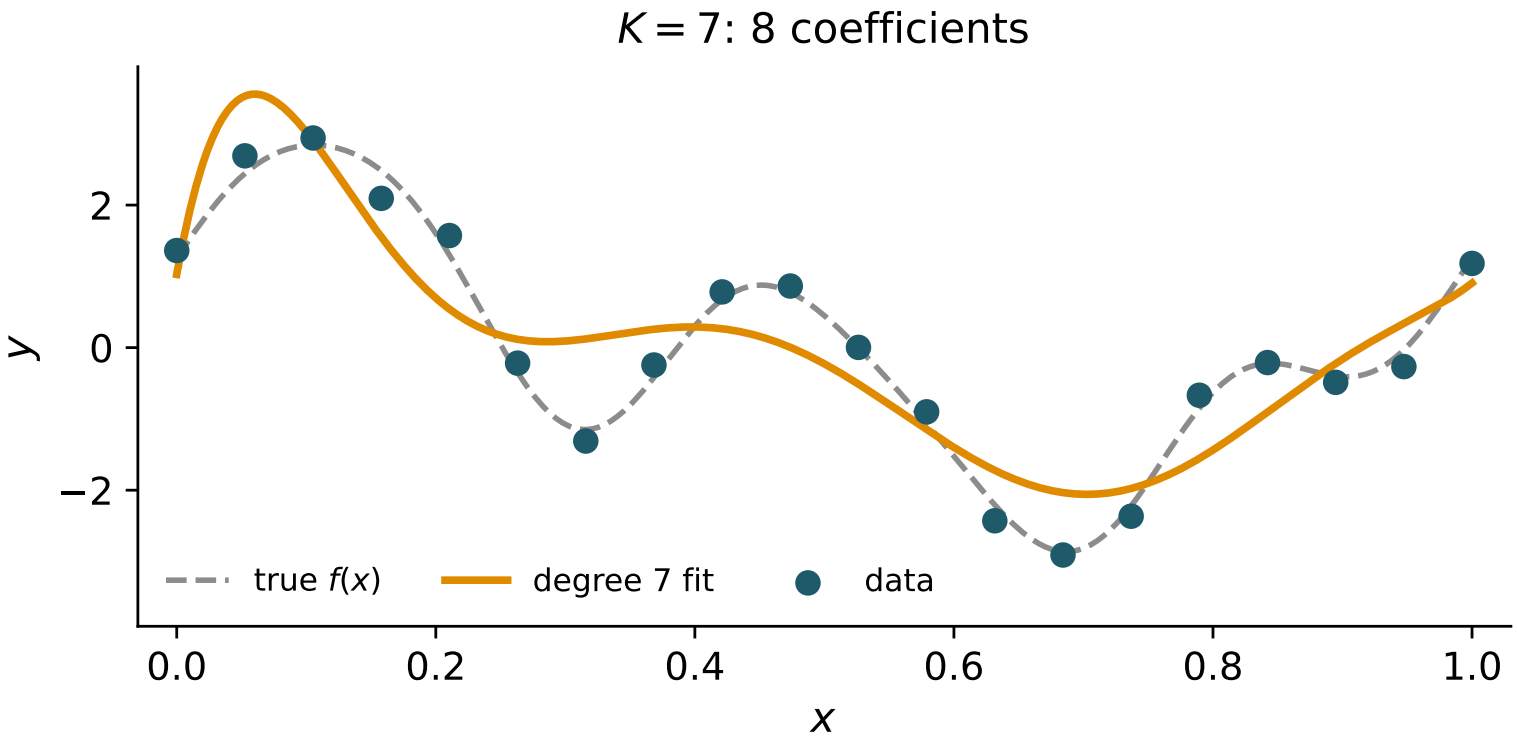
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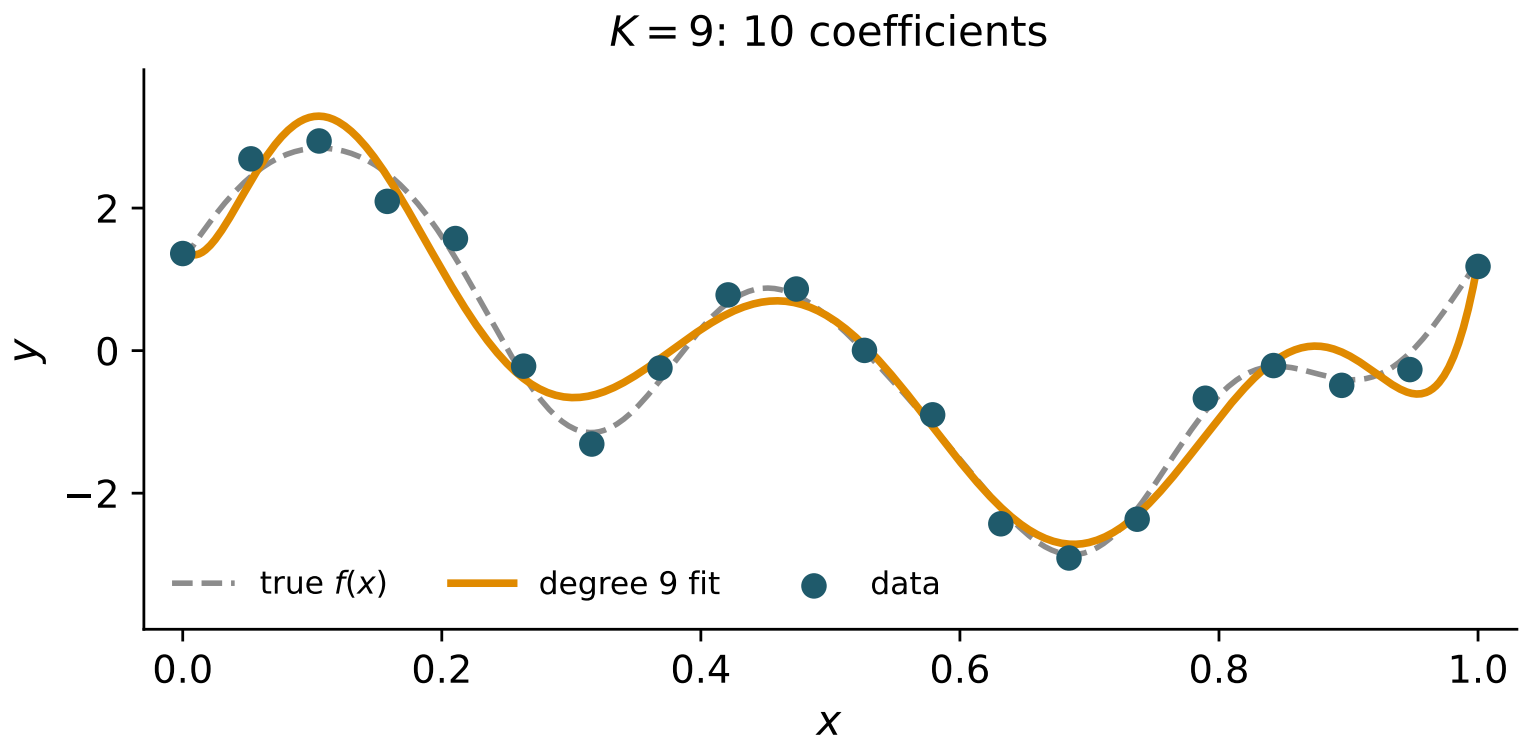
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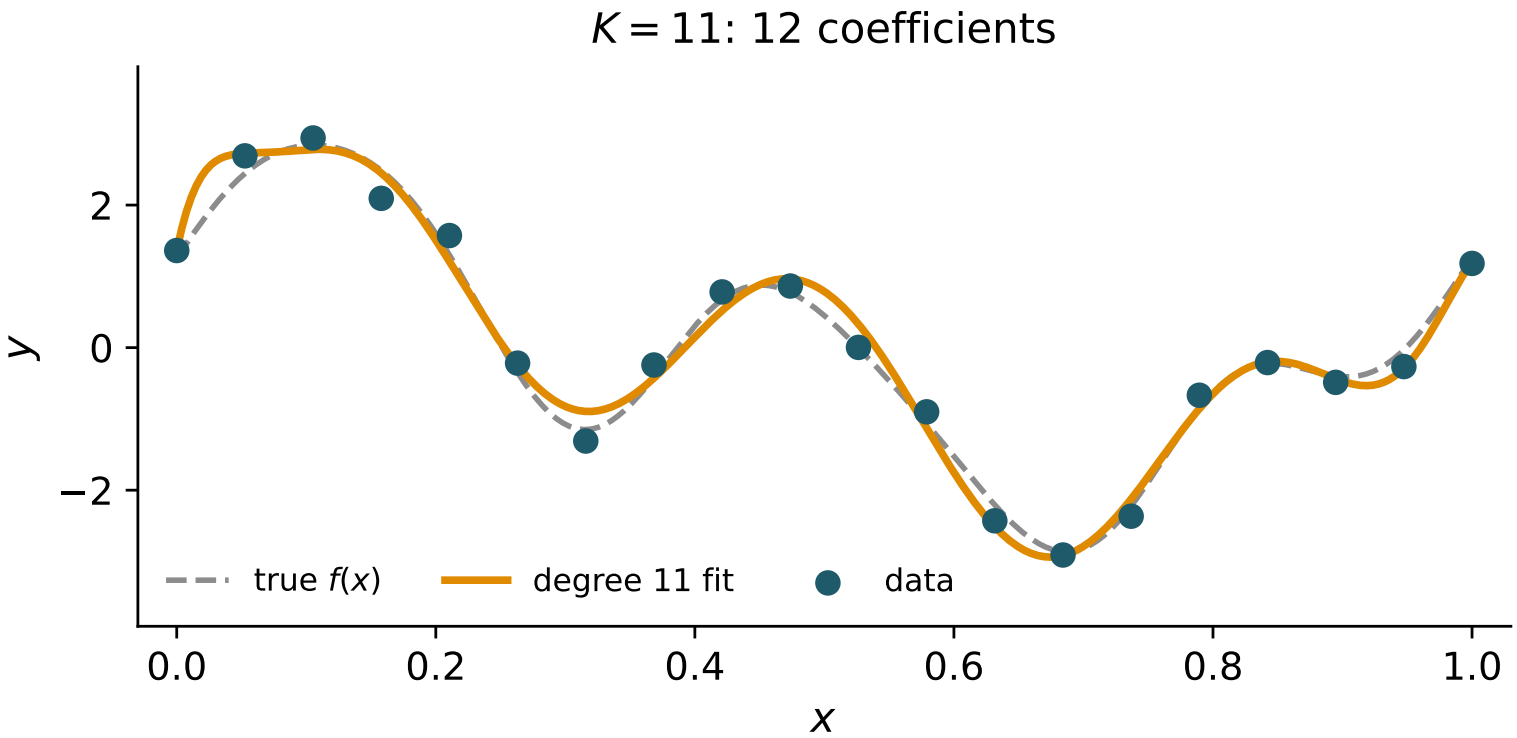
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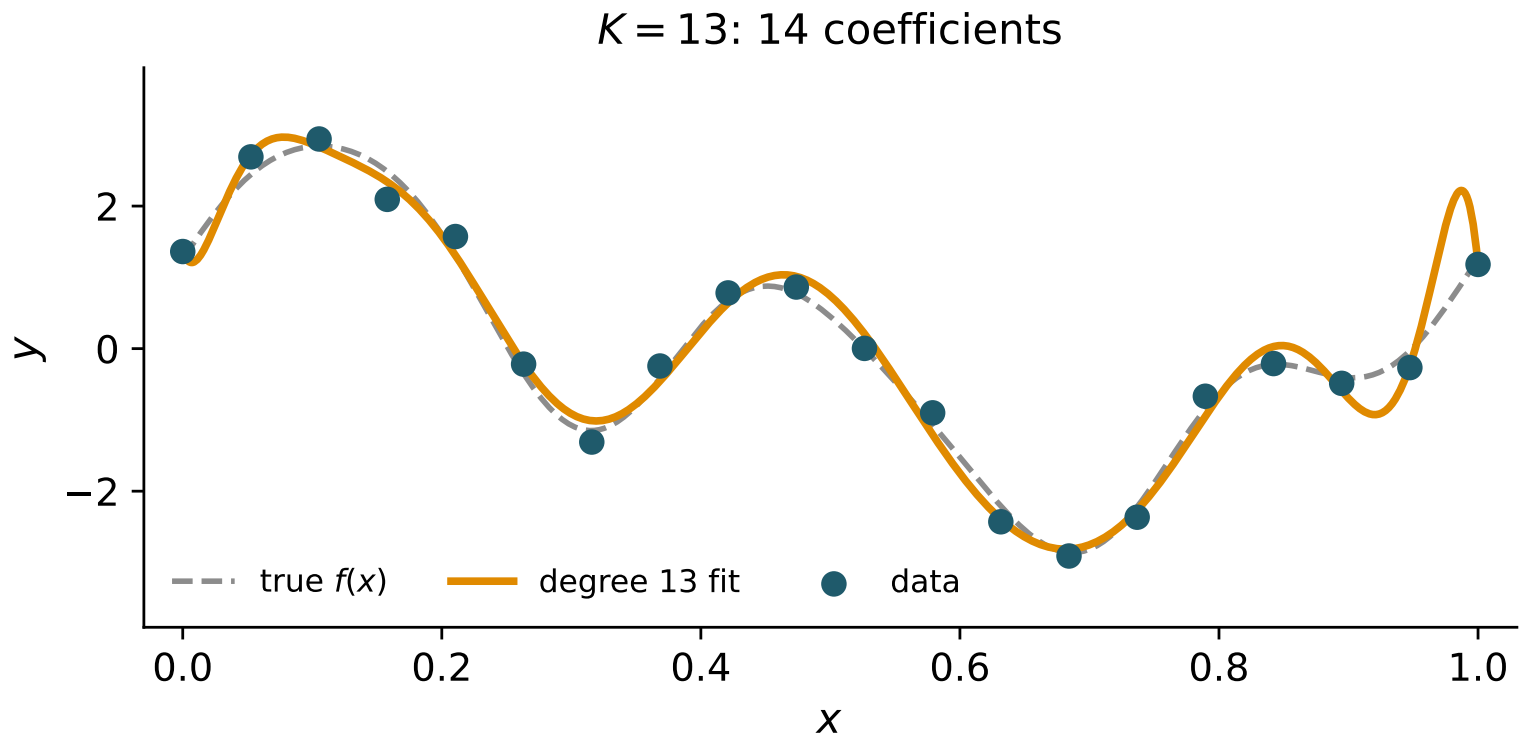
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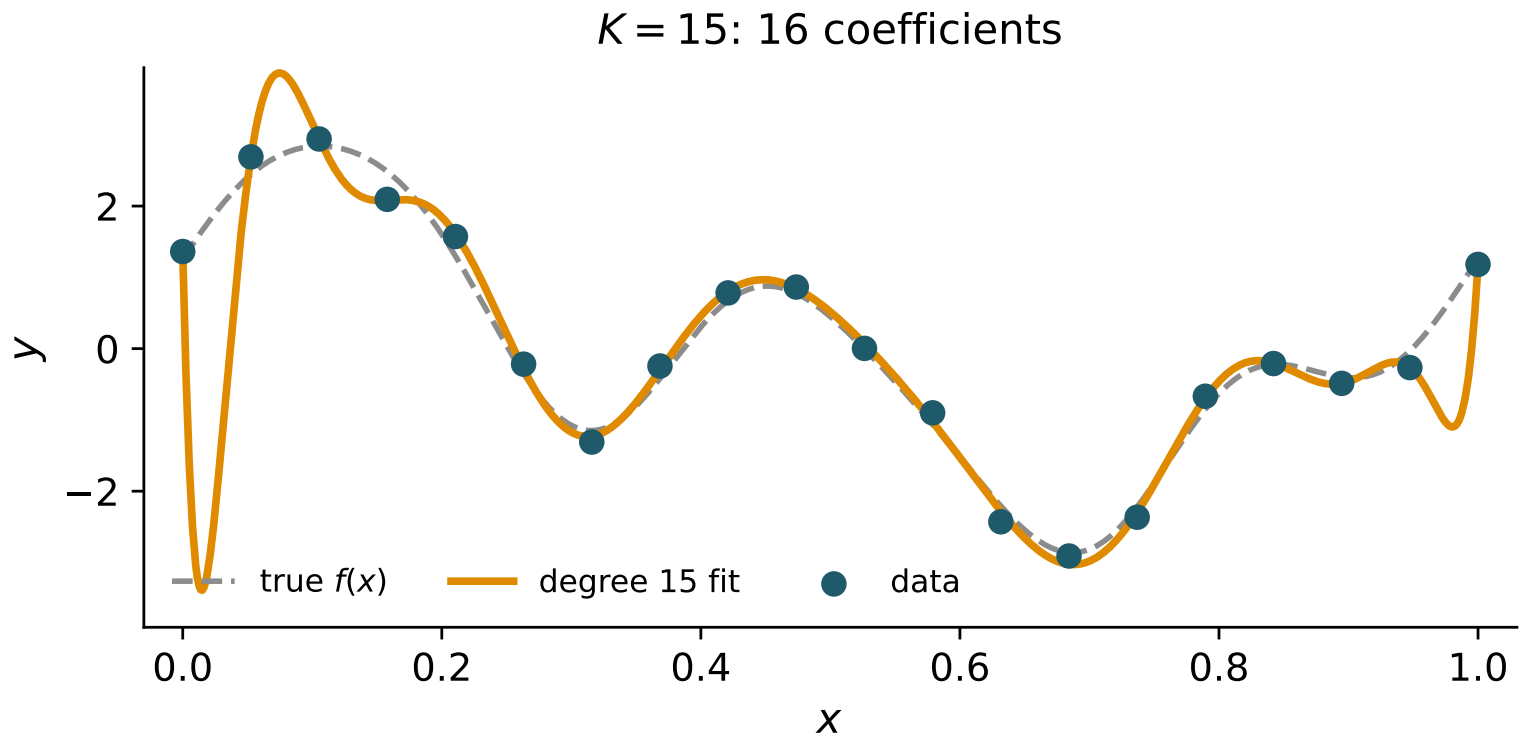
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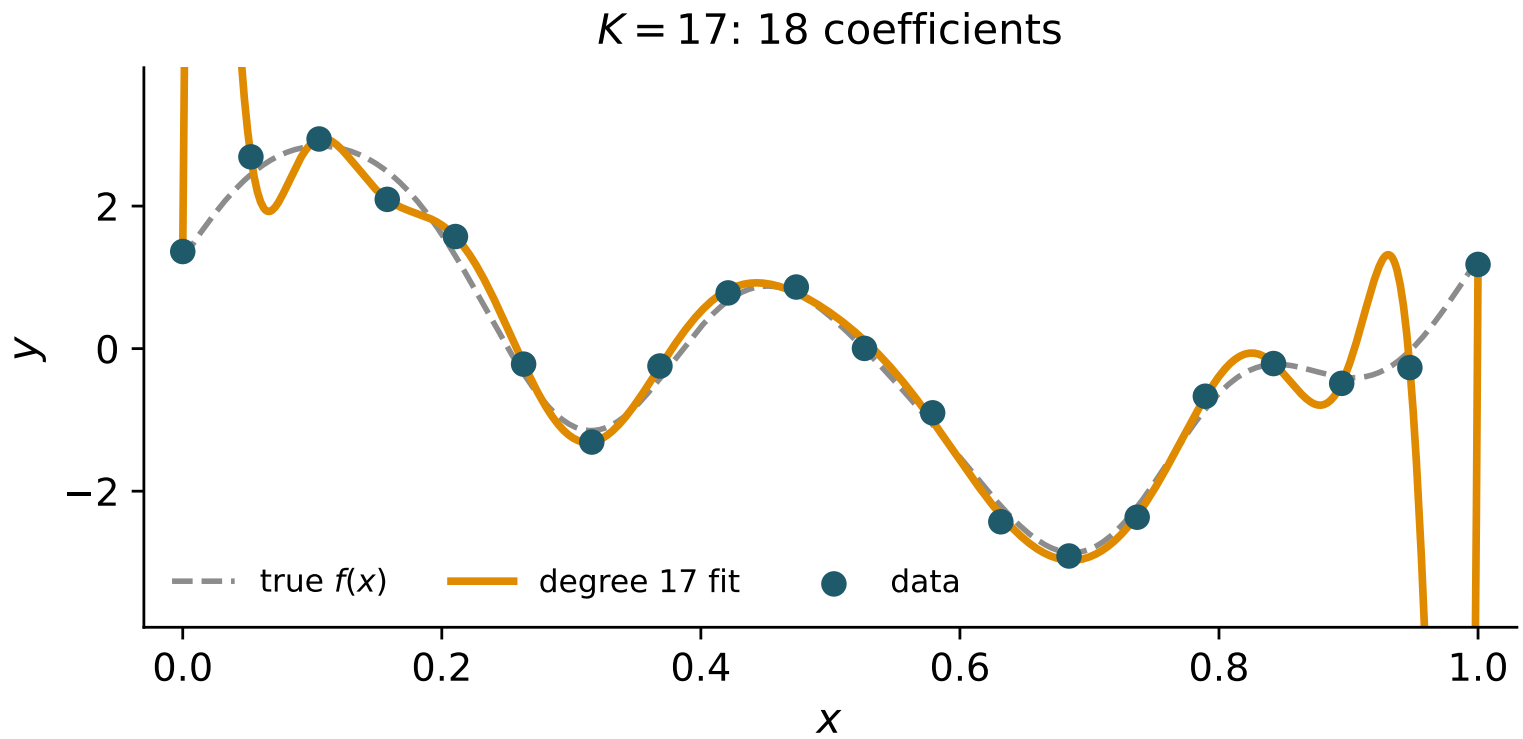
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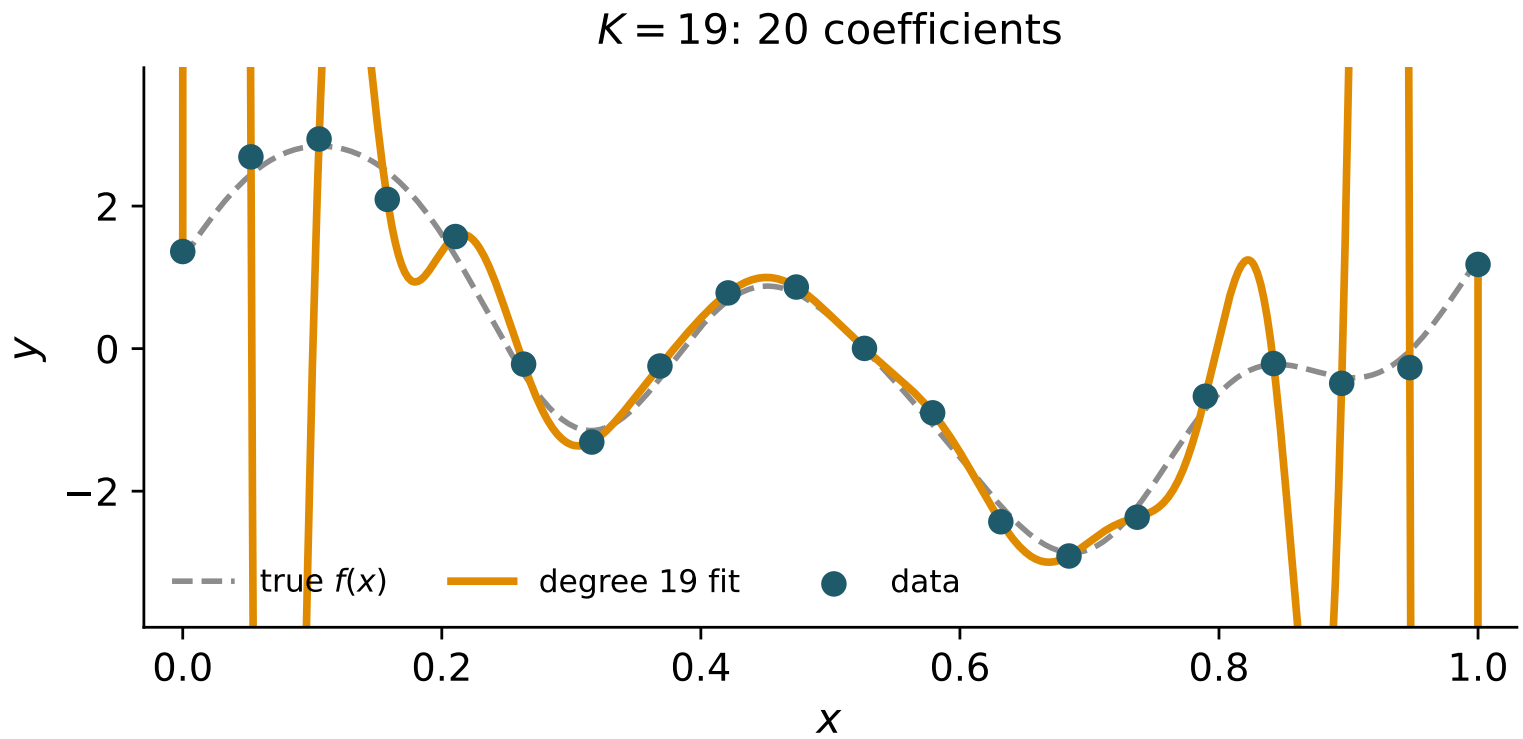
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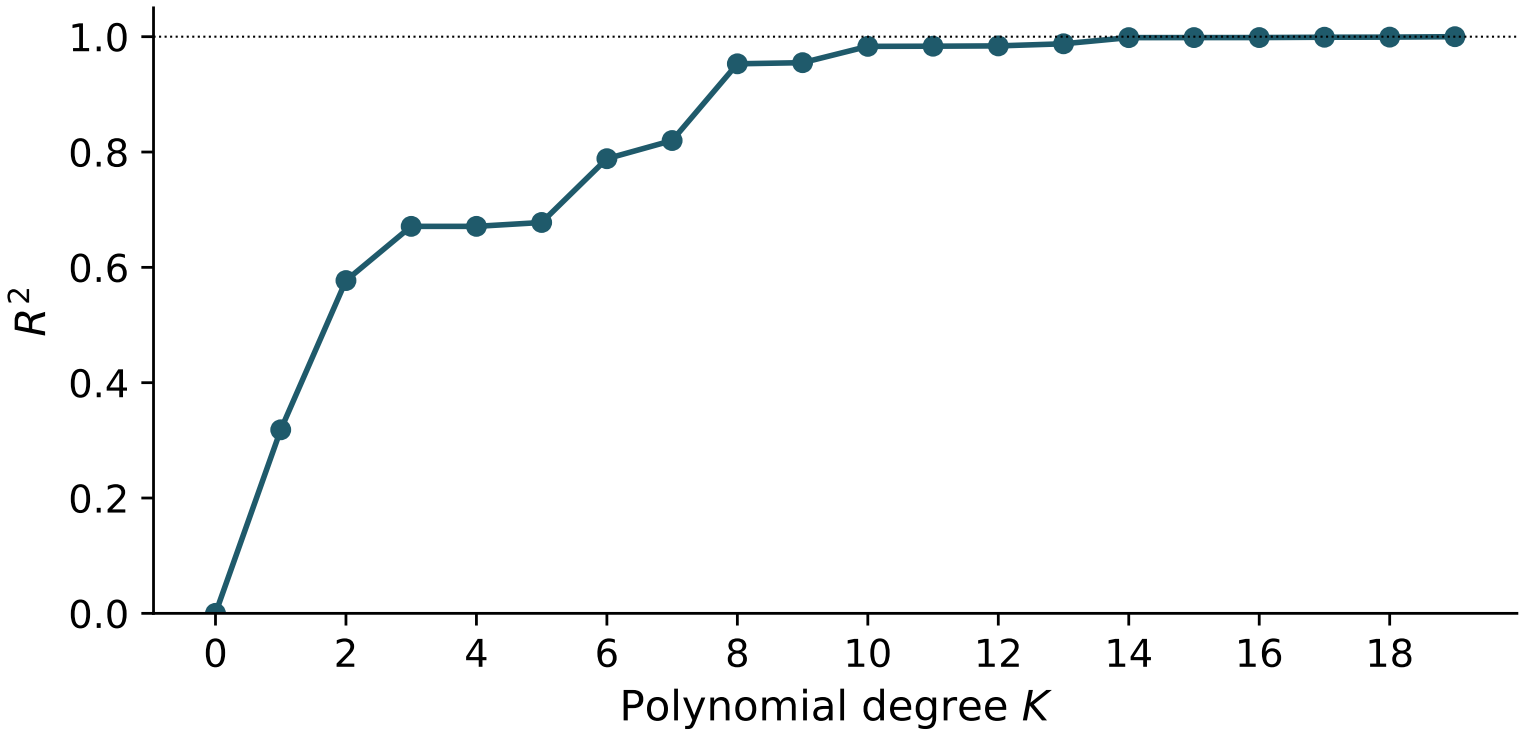


Example: polynomials fitted to a random function



R^2 as the degree increases

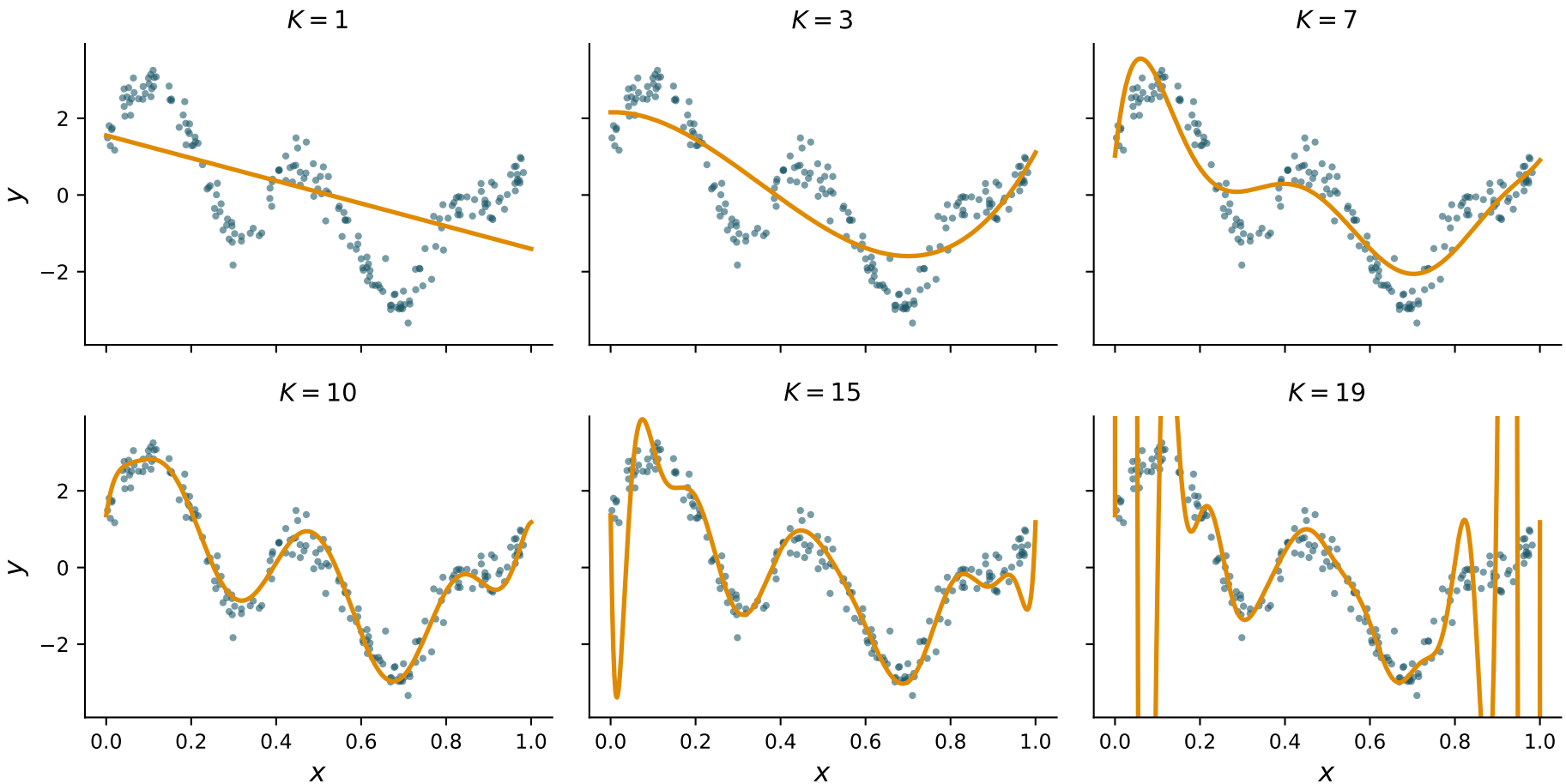
R^2 for the same 20 points and every degree $K = 0, 1, \ldots, 19$:



R^2 goes up with every coefficient we add and reaches 1 at $K = 19$, where the polynomial passes through all 20 points.

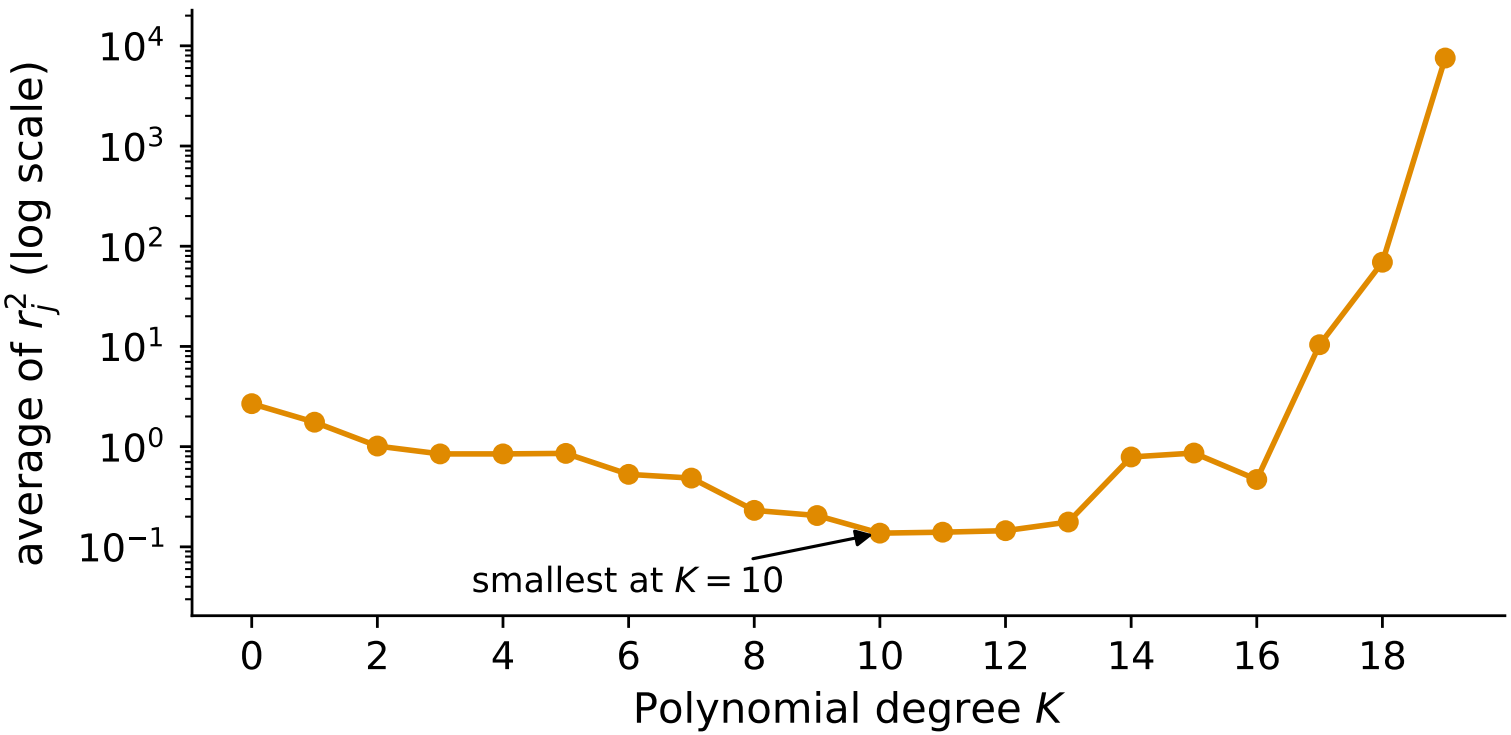
New points and the fitted polynomials

200 new random points from the same function, with the same noise, together with the polynomials fitted to the original 20 points. The residual $r_j = y_j - \hat{y}_j$ of a new point is its vertical distance from the curve.



Average squared residual of the new points

For each degree K, the average of r_j^2 over the 200 new points:



It is smallest at $K = 10$. For larger K it grows quickly, and at $K = 19$ it is about 7500.